

Q1: LED is constructed from Ga-As-P semiconducting material. The energy gap of this LED is 1.9 eV. Calculate the wavelength of light emitted and its colour.

[$h = 6.63 \times 10^{-34}$ Js and $c = 3 \times 10^8$ ms⁻¹]

- (A) 654 nm and red colour
- (B) 1046 nm and blue colour
- (C) 1046 nm and red colour
- (D) 654 nm and orange colour

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Solution:

$$\lambda = \frac{hc}{E} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.9 \times 1.6 \times 10^{-19}} = 6.54 \times 10^{-7}$$

= 654 nm
Red color

Q2: Assume that a tunnel is dug along a chord of the earth, at a perpendicular distance ($R/2$) from the earth's centre, where 'R' is the radius of the Earth. The wall of the tunnel is frictionless. If a particle is released in this tunnel, it will execute a simple harmonic motion with a time period:

(A) $\frac{1}{2\pi} \sqrt{\frac{g}{R}}$

(B) $2\pi \sqrt{\frac{R}{g}}$

(C) $\frac{2\pi R}{g}$

(D) $\frac{g}{2\pi R}$

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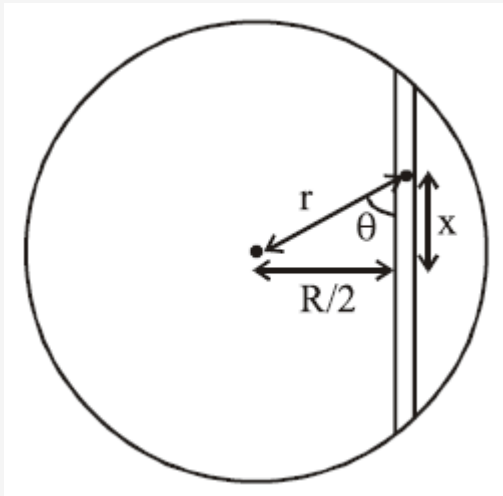
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Solution:



Force along the tunnel

$$F = - \left(\frac{GMmr}{R^3} \right) \cos \theta$$

$$F = - \frac{gm}{R} x \left(\frac{GM}{R^2} = g, r \cos \theta = x \right)$$

$$a = - \frac{g}{R} x$$

$$\omega^2 = \frac{g}{R} \quad T = 2\pi \sqrt{\frac{R}{g}}$$

Q3: If λ_1 and λ_2 are the wavelengths of the third member of Lyman and first member of the Paschen series respectively, then the value of $\lambda_1 : \lambda_2$ is :

- (A) 7 : 108
- (B) 1 : 9
- (C) 1 : 3
- (D) 7 : 135

Q3: If λ_1 and λ_2 are the wavelengths of the third member of Lyman and first member of the Paschen series respectively, then the value of $\lambda_1 : \lambda_2$ is :

(A) 7 : 108

(B) 1 : 9

(C) 1 : 3

(D) 7 : 135

Solution:

$$\frac{1}{\lambda_1} = R \left[\frac{1}{1^2} - \frac{1}{4^2} \right]$$

$$\frac{1}{\lambda_2} = R \left[\frac{1}{3^2} - \frac{1}{4^2} \right]$$

$$\frac{\lambda_1}{\lambda_2} = \frac{\left[\frac{1}{9} - \frac{1}{16} \right]}{\left[1 - \frac{1}{16} \right]} = \frac{7}{9 \times 15}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{7}{135}$$

Q4: A particle is moving with uniform speed along the circumference of a circle of radius R under the action of a central fictitious force F which is inversely proportional to R^3 . Its time period of revolution will be given by:

(A) $T \propto R^{\frac{4}{3}}$

(B) $T \propto R^2$

(C) $T \propto R^{\frac{5}{2}}$

(D) $T \propto R^{\frac{3}{2}}$

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Solution:

$$\begin{aligned} F &\propto \frac{1}{R^3} \\ \frac{K}{R^3} &= m\omega^2 R \\ \omega^2 &= \frac{K}{m} \times \frac{1}{R^4} \\ \left(\frac{2\pi}{T}\right)^2 &= \frac{K}{m} \times \frac{1}{R^4} \\ T^2 &\propto R^4 \\ T &\propto R^2 \end{aligned}$$

Q5: A planet revolving in elliptical orbit has :

- (A) a constant velocity of revolution.
- (B) has the least velocity when it is nearest to the sun.
- (C) its areal velocity is directly proportional to its velocity.
- (D) areal velocity is inversely proportional to its velocity.
- (E) to follow a trajectory such that the areal velocity is constant.

Choose the correct answer from the options given below :

- (A) A only
- (B) C only
- (C) E only
- (D) D only

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- (A) A only
- (B) C only
- ☒ (C) E only
- (D) D only

Solution:

As per Kepler's 2nd law, Areal velocity is constant.

Q6: Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Body 'P' having mass M moving with speed ' u ' has head-on collision elastically with another body 'Q' having mass ' m ' initially at rest. If $m \ll M$, body 'Q' will have a maximum speed equal to ' $2u$ ' after collision.

Reason R : During elastic collision, the momentum and kinetic energy are both conserved. In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Both A and R are correct but R is NOT the correct explanation of A
- (B) Both A and R are correct and R is the correct explanation of A
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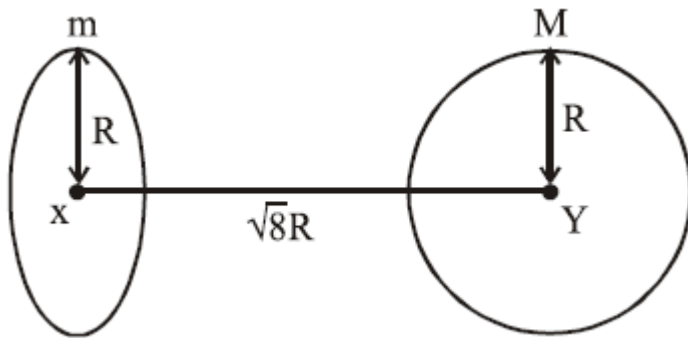
Solution:

For $e = 1$ & second body at rest

$$V_2 = \frac{2m_1 u_1}{m_1 + m_2} = \frac{2u(M)}{M+m} \simeq 2u$$

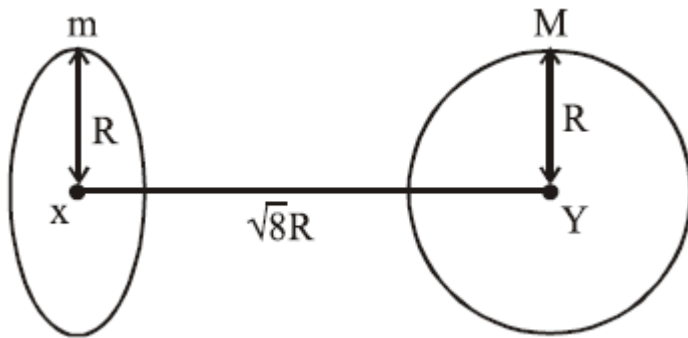
Since $M \gg m$

Q7: Find the gravitational force of attraction between the ring and sphere as shown in the diagram, where the plane of the ring is perpendicular to the line joining the centres. If $\sqrt{8}R$ is the distance between the centres of a ring (of mass 'm') and a sphere (mass 'M') where both have equal radius 'R'.



- (A) $\frac{\sqrt{8}}{27} \cdot \frac{GmM}{R^2}$
- (B) $\frac{1}{3\sqrt{8}} \cdot \frac{GMm}{R^2}$
- (C) $\frac{2\sqrt{2}}{3} \cdot \frac{GMm}{R^2}$
- (D) $\frac{\sqrt{8}}{9} \cdot \frac{GmM}{R}$

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Solution:

Gravitational field of ring

$$= - \frac{Gm\mathbf{x}}{(R^2 + x^2)^{3/2}}$$

Force between sphere & ring

$$= \frac{GmM(\sqrt{8}R)}{(R^2 + 8R^2)^{3/2}}$$

$$= \frac{GmM}{R^2} \times \frac{\sqrt{8}}{27}$$

Q8: Four identical solid spheres each of mass 'm' and radius 'a' are placed with their centres on the four corners of a square of side 'b'. The moment of inertia of the system about one side of square where the axis of rotation is parallel to the plane of the square is

(A) $\frac{4}{5}ma^2 + 2mb^2$

(B) $\frac{8}{5}ma^2 + mb^2$

(C) $\frac{8}{5}ma^2 + 2mb^2$

(D) $\frac{4}{5}ma^2$

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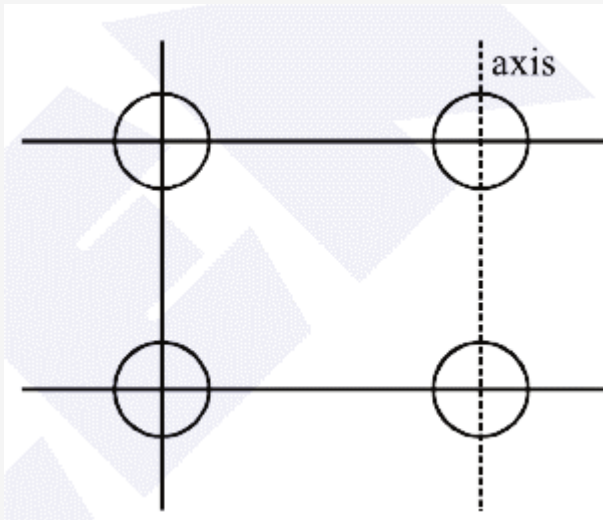
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(B) $\frac{8}{5}ma^2 + mb^2$

(C) $\frac{8}{5}ma^2 + 2mb^2$

(D) $\frac{4}{5}ma^2$

Solution:



$$I = 2 \times \left(\frac{2}{5} ma^2 \right) + 2 \times \left(\frac{2}{5} ma^2 + mb^2 \right)$$

$$I = \frac{8}{5} ma^2 + 2mb^2$$

Q9: A large number of water drops, each of radius r , combine to have a drop of radius R . If the surface tension is T and mechanical equivalent of heat is J , the rise in heat energy per unit volume will be :

(A) $\frac{2T}{rJ}$

(B) $\frac{2T}{J} \left(\frac{1}{r} \times \frac{1}{R} \right)$

(C) $\frac{3T}{J} \left(\frac{1}{r} - \frac{1}{R} \right)$

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Solution:

$$n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$$

$$\therefore n^{1/3} r = R$$

$\therefore$ Total change in surface energy

$$= (n(4\pi r^2) - 4\pi R^2) T$$

$$\Rightarrow 4\pi T (nr^2 - R^2)$$

$\therefore$ Heat energy

$$= \frac{4\pi T(nr^2 - R^2)}{J \times \frac{4}{3}\pi R^3} = \frac{3T}{J} \left(\frac{nr^2}{R^3} - \frac{1}{R} \right)$$

$$\text{Put } nr^3 = R^3$$

$$\therefore \frac{3T}{J} \left(\frac{1}{r} - \frac{1}{R} \right)$$

Q10: An alternating current is given by the equation $i = i_1 \sin \omega t + i_2 \cos \omega t$. The rms current will be :

(A) $\frac{1}{\sqrt{2}} (i_1^2 + i_2^2)^{\frac{1}{2}}$

(B) $\frac{1}{\sqrt{2}} (i_1 + i_2)^2$

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Solution:

$$i = i_1 \sin \omega t + i_2 \sin(\omega t + 90)$$

$$i = \sqrt{i_1^2 + i_2^2} \sin(\omega t + \phi)$$

$$i_{rms} = \frac{i_0}{\sqrt{2}} = \frac{\sqrt{i_1^2 + i_2^2}}{\sqrt{2}}$$

Q11: The normal density of a material is ρ and its bulk modulus of elasticity is K . The magnitude of increase in density of material, when a pressure P is applied uniformly on all sides, will be :

(A) $\frac{PK}{\rho}$

(B) $\frac{K}{\rho P}$

(C) $\frac{\rho K}{P}$

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Solution:

$$\rho = \frac{M}{V}$$

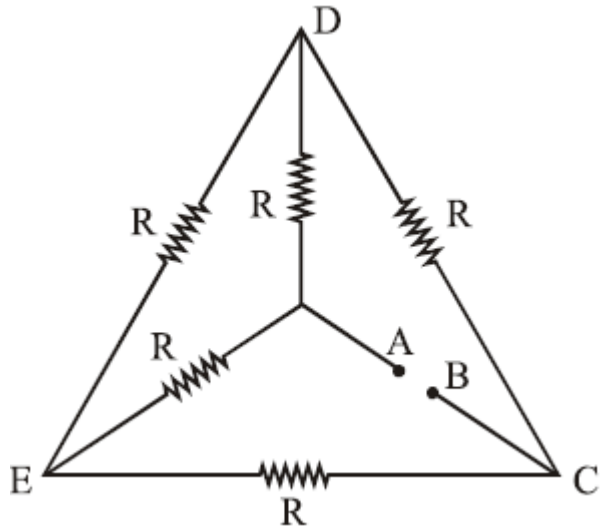
$$\frac{d\rho}{\rho} = -\frac{dV}{V}$$

$$K = -\frac{P}{\frac{dV}{V}}$$

$$-\frac{dV}{V} = \frac{P}{K}$$

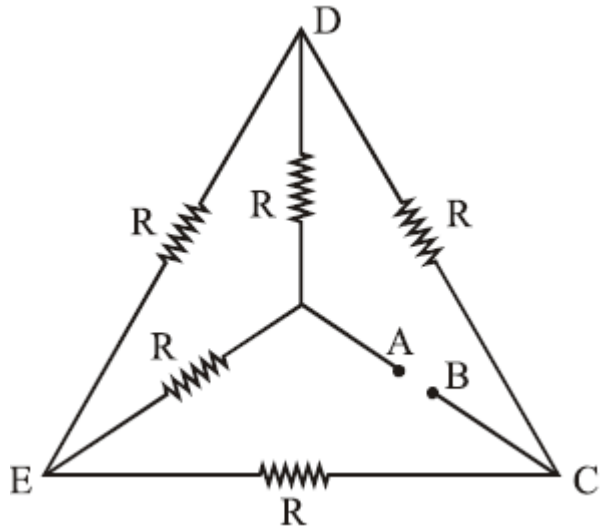
$$\frac{d\rho}{\rho} = \frac{P}{K} \Rightarrow d\rho = \frac{\rho P}{K}$$

Q12: Five equal resistances are connected in a network as shown in figure. The net resistance between the points A and B is



- (A) $\frac{3R}{2}$
- (B) $\frac{R}{2}$
- (C) R
- (D) $2R$

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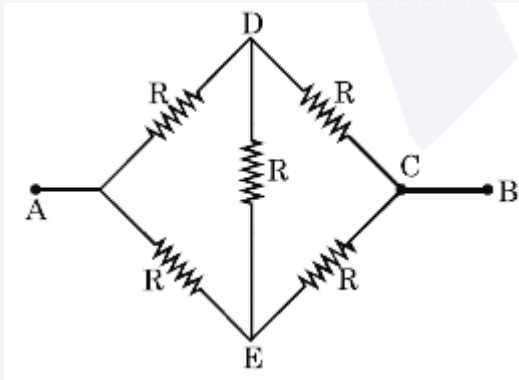
(B) $\frac{R}{2}$

(C) R

(D) $2R$

Solution:

The diagram can be drawn like



It is a wheat stone bridge

$$\therefore R_{eq} = \frac{2R \times 2R}{2R + 2R} \Rightarrow R$$

Q13: If two similar springs each of spring constant K_1 are joined in series, the new spring constant and time period would be changed by a factor :

(A) $\frac{1}{2}, 2\sqrt{2}$

(B) $\frac{1}{4}, \sqrt{2}$

(C) $\frac{1}{4}, 2\sqrt{2}$

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Solution:

$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$
$$\frac{1}{k_{eq}} = \frac{1}{k} + \frac{1}{k} \Rightarrow k_{eq} = \frac{k}{2}$$

$$k' = \frac{k}{2}$$

$$T = 2\pi\sqrt{\frac{M}{m}} \quad T' = 2\pi\sqrt{\frac{M}{k'}}$$

$$\Rightarrow T' = 2\pi\sqrt{\frac{M}{\frac{k}{2}}} \times \sqrt{2}$$

$$T' = \sqrt{2}T$$

Q14: A short straight object of height 100 cm lies before the central axis of a spherical mirror whose focal length has absolute value $|f| = 40$ cm. The image of object produced by the mirror is of height 25 cm and has the same orientation of the object. One may conclude from the information :

- (A) Image is virtual, opposite side of convex mirror
- (B) Image is real, same side of concave mirror
- (C) Image is virtual, opposite side of concave mirror
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Solution:

Since orientation is same image is virtual. Since image is smaller the mirror has to be convex.

Q15: In a Young's double slit experiment two slits are separated by 2 mm and the screen is placed one meter away. When a light of wavelength 500 nm is used, the fringe separation will be :

- (A) 0.75 mm
- (B) 0.25 mm
- (C) 0.50 mm
- (D) 1 mm

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(D) 1 mm

Solution:

$$\beta = \frac{\lambda D}{d} = \frac{500 \times 10^{-9} \times 1}{2 \times 10^{-3}}$$

$$\beta = \frac{5}{2} \times 10^{-4} m = 2.5 \times 10^{-1} mm$$

$$b = 0.25 \text{ mm}$$

Q16: Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : An electron microscope can achieve better resolving power than an optical microscope.

Reason R : The de Broglie's wavelength of the electrons emitted from an electron gun is much less than wavelength of visible light.

In the light of the above statements, choose the correct answer from the options given below:

- (A) Both A and R are true but R is NOT the correct explanation of A
- (B) A is false but R is true.
- (C) A is true but R is false.
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Solution:

$$\text{Resolving power} \propto \frac{1}{\lambda}$$

Since wavelength of electron is much less than visible light, its resolving power will be much more.

Q17: In a typical combustion engine the work done by a gas molecule is given by

$W = \alpha^2 \beta e^{\frac{-\beta x^2}{kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. If α and β are constants, dimensions of α will be :

- (A) $[M^0 L T^0]$
- (B) $[M^2 L T^{-2}]$
- (C) $[M L T^{-2}]$
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(B) $[M^2 L T^{-2}]$

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(D) $[M L T^{-1}]$

Solution:

kT has dimension of energy

$\frac{\beta x^2}{kT}$ is dimensionless

$$[\beta] [L^2] = [M L^2 T^{-2}]$$

$$[\beta] = [M T^{-2}]$$

$\alpha^2 \beta$ has dimensions of work

$$[\alpha^2] [M T^{-2}] = [M L^2 T^{-2}]$$

$$[\alpha] = [M^0 L T^0]$$

Q18: Consider the combination of 2 capacitors C_1 and C_2 with $C_2 > C_1$ when connected in parallel, the equivalent capacitance is $\frac{15}{4}$ times the equivalent capacitance of the same connected in series. Calculate the ratio of capacitors $\frac{C_2}{C_1}$.

(A) $\frac{15}{11}$

(B) $\frac{15}{4}$

(C) $\frac{29}{15}$

(D) $\frac{111}{80}$

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(B) $\frac{15}{4}$

(C) $\frac{29}{15}$

(D) $\frac{111}{80}$

Solution:

When connected in parallel

$$C_{eq} = C_1 + C_2$$

When in series

$$C'_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

$$C_1 + C_2 = \frac{15}{4} \left(\frac{C_1 C_2}{C_1 + C_2} \right)$$

$$4(C_1 + C_2)^2 = 15 C_1 C_2$$

$$4C_1^2 + 4C_2^2 - 7C_1 C_2 = 0$$

$$\text{dividing by } C_1^2 \quad 4\left(\frac{C_2}{C_1}\right)^2 - \frac{7C_2}{C_1} + 4 = 0$$

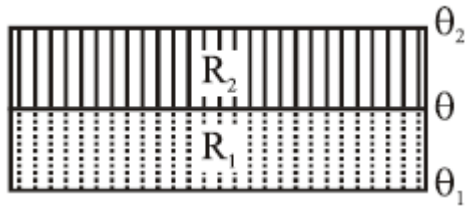
$$\text{Let } \frac{C_2}{C_1} = x$$

$$4x^2 - 7x + 4 = 0$$

$$b^2 - 4ac = 49 - 64 < 0$$

No solution exists

Q19: The temperature θ at the junction of two insulating sheets, having thermal resistances R_1 and R_2 as well as top and bottom temperatures θ_1 and θ_2 (as shown in figure) is given by :



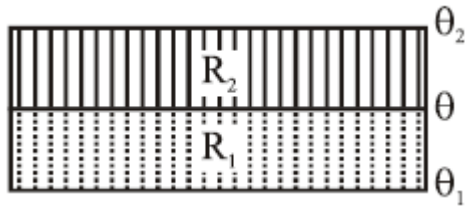
(A) $\frac{\theta_2 R_2 - \theta_1 R_1}{R_2 - R_1}$

(B) $\frac{\theta_1 R_2 + \theta_2 R_1}{R_1 + R_2}$

(C) $\frac{\theta_1 R_1 + \theta_2 R_2}{R_1 + R_2}$

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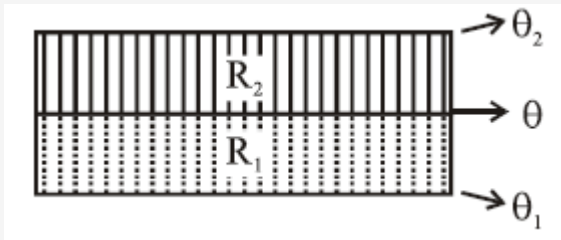
(A) $\frac{\theta_2 R_2 - \theta_1 R_1}{R_2 - R_1}$

(B) $\frac{\theta_1 R_2 + \theta_2 R_1}{R_1 + R_2}$

(C) $\frac{\theta_1 R_1 + \theta_2 R_2}{R_1 + R_2}$

(D) $\frac{\theta_1 R_2 - \theta_2 R_1}{R_2 - R_1}$

Solution:



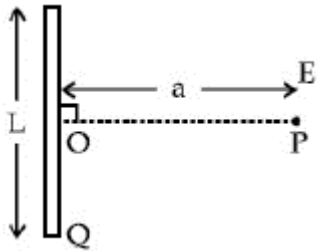
Heat flow rate will be same through both

$$\therefore \frac{\theta_1 - \theta}{R_1} = \frac{\theta - \theta_2}{R_2}$$

$$R_2 \theta_1 - R_2 \theta = R_1 \theta - R_1 \theta_2$$

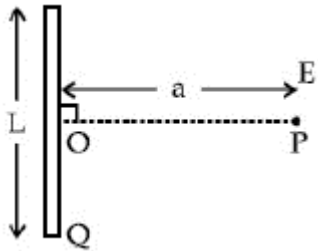
$$\theta = \frac{R_2 \theta_1 + R_1 \theta_2}{R_1 + R_2}$$

Q20: Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q . The distance of the point P from the centre of the rod is $a = \frac{\sqrt{3}}{2}L$



- (A) $\frac{Q}{4\pi\epsilon_0 L^2}$
- (B) $\frac{\sqrt{3}Q}{4\pi\epsilon_0 L^2}$
- (C) $\frac{Q}{3\pi\epsilon_0 L^2}$
- (D) $\frac{Q}{2\sqrt{3}\pi\epsilon_0 L^2}$

Q20: Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q. The distance of the point P from the centre of the rod is $a = \frac{\sqrt{3}}{2}L$



(A) $\frac{Q}{4\pi\epsilon_0 L^2}$

(B) $\frac{\sqrt{3}Q}{4\pi\epsilon_0 L^2}$

(C) $\frac{Q}{3\pi\epsilon_0 L^2}$

(D) $\frac{Q}{2\sqrt{3}\pi\epsilon_0 L^2}$

Solution:

$$E = \frac{k\lambda}{a} (\sin \theta_1 + \sin \theta_2)$$

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{Q}{L} \times \frac{1}{\left(\frac{\sqrt{3}L}{2}\right)} \times (2 \sin \theta)$$

$$\tan \theta = \frac{L/2}{\frac{\sqrt{3}L}{2}} = \frac{1}{\sqrt{3}}$$

$$\sin \theta = \frac{1}{2}$$

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{2Q}{\sqrt{3}L^2} \times \left(2 \times \frac{1}{2}\right)$$

$$E = \frac{Q}{2\sqrt{3}\pi\epsilon_0 L^2}$$

Q21: A radiation is emitted by 1000 W bulb and it generates an electric field and magnetic field at P, placed at a distance of 2 m. The efficiency of the bulb is 1.25%. The value of peak electric field at P is $x \times 10^{-1}$ V/m. Value of x is _____. (Rounded-off to the nearest integer)
[Take $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$, $c = 3 \times 10^8 \text{ ms}^{-1}$]

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 [Take $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$, $c = 3 \times 10^8 \text{ ms}^{-1}$]

137.00

Solution:

$$I_{avg} = \frac{1}{2} \epsilon_0 E_0^2 C$$

$$\frac{1.25}{100} \times \frac{1000}{4\pi(2)^2} = \frac{1}{2} \times 8.85 \times 10^{-12} \times 3 \times 10^8 \times E_0^2$$

$$E_0^2 = 187.4$$

$$\therefore E_0 = 13.689 \text{ V/m}$$

$$= 136.89 \times 10^{-1} \text{ V/m}$$

$$\therefore x = 136.89$$

Rounding off to nearest integer

$$x = 137$$

Q22: The mass per unit length of a uniform wire is 0.135 g/cm. A transverse wave of the form $y = -0.21 \sin(x + 30t)$ is produced in it, where x is in meter and t is in second. Then, the expected value of tension in the wire is $x \times 10^{-2}$ N. N. Value of x is _____
(Round-off to the nearest integer)

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(Round-off to the nearest integer)

1215.00

Solution:

$$\mu = 0.135 \text{ gm/cm} = 0.0135 \text{ kg/m}$$

$$y = -0.21 \sin(x + 30t)$$

(x in meter & t in sec)

$$v = \frac{\omega}{k} = \frac{30}{1} = 30 \text{ m/s}$$

$$v = \sqrt{\frac{T}{\mu}} \Rightarrow T = v^2 \mu = (30)^2 (0.0135)$$

$$= 12.15$$

$$= x \times 10^{-2} N$$

$$\Rightarrow x = 1215$$

Q23: A container is divided into two chambers by a partition. The volume of first chamber is 4.5 litre and second chamber is 5.5 litre. The first chamber contain 3.0 moles of gas at pressure 2.0 atm and second chamber contain 4.0 moles of gas at pressure 3.0 atm. After the partition is removed and the mixture attains equilibrium, then, the common equilibrium pressure existing in the mixture is $x \times 10^{-1}$ atm. Value of x is _____.

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26.00

Solution:

Let common equilibrium pressure of mixture is P atmp, then

$$U_1 + U_2 = U_{\text{mixutre}}$$

$$\frac{f}{2} P_1 V_1 + \frac{f}{2} P_2 V_2 = \frac{f}{2} P (V_1 + V_2)$$

$$\frac{f}{2} (2) (4.5) + \frac{f}{2} (3) (5.5) = \frac{f}{2} P (4.5 + 5.5)$$

$$\Rightarrow P = 2.55 = x \times 10^{-1} \text{ atmp}$$

So, $x = 25.5 \approx 26$ (Nearest integer)

Q24: The maximum and minimum amplitude of an amplitude modulated wave is 16 V and 8 V respectively. The modulation index for this amplitude modulated wave is $x \times 10^{-2}$. The value of x is _____

Q24: The maximum and minimum amplitude of an amplitude modulated wave is 16 V and 8 V respectively. The modulation index for this amplitude modulated wave is $x \times 10^{-2}$. The value of x is _____

33.00

Solution:

$$\text{Modulation index} = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$$

$$= \frac{16-8}{16+8} = \frac{8}{24} = \frac{1}{3} = 0.33$$

$$x \times 10^{-2} = 0.33$$

$$x = 33$$

Q25: In an electrical circuit, a battery is connected to pass 20 C of charge through it in a certain given time. The potential difference between two plates of the battery is maintained at 15 V. The work done by the battery is _____ J.

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Solution:

$$\text{Work done by battery} = Q (\Delta V)$$

$$\Rightarrow 20 \times 15 = 300 \text{ J}$$

$$\therefore 300$$

Q26:

In a series LCR resonant circuit, the quality factor is measured as 100. If the inductance is increased by two fold and resistance is decreased by two fold, then the quality factor after this change will be _____. (Take $\sqrt{2} = 1.414$)

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282.84

Solution:

$$Q = \frac{X_L}{R} = \frac{\omega L}{R} = \frac{1}{\sqrt{LC}} \frac{L}{R} = \frac{\sqrt{L}}{R\sqrt{C}}$$

$$Q' = \frac{\sqrt{2L}}{\left(\frac{R}{2}\right)\sqrt{C}} = 2\sqrt{2}Q = 2\sqrt{2}(100) \\ = 282.84$$

Q27: A boy pushes a box of mass 2 kg with a force $\vec{F} = (20\hat{i} + 10\hat{j}) \text{ N}$ on a frictionless surface. If the box was initially at rest, then _____ m is displacement along the x-axis after 10 s.

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Solution:

$$\vec{F} = 20\hat{i} + 10\hat{j}$$

$$\vec{a} = \frac{\vec{F}}{m} = \frac{20\hat{i} + 10\hat{j}}{2} \Rightarrow 10\hat{i} + 5\hat{j}$$

$$\therefore \vec{s} = \frac{1}{2}\vec{a}t^2 = \frac{1}{2}(10\hat{i} + 5\hat{j}) \times (10)^2$$

$$\Rightarrow 50(10\hat{i} + 5\hat{j}) \text{ m}$$

$\therefore$ Displacement along x-axis

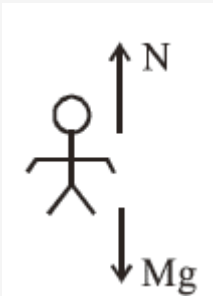
$$\Rightarrow 50 \times 10 \Rightarrow 500 \text{ m}$$

$$\therefore 500$$

Q28: A person standing on a spring balance inside a stationary lift measures 60 kg. The weight of that person if the lift descends with uniform downward acceleration of 1.8 m/s^2 will be _____ N. [$g = 10 \text{ m/s}^2$]

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Solution:

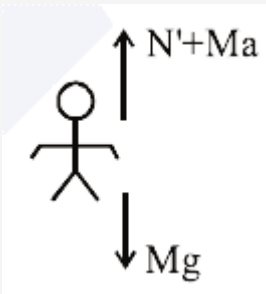


When lift is at rest

$$N = mg$$

$$\Rightarrow 60 \times 10 = 600N$$

When lift moves with downward acceleration. In frame of lift pseudo force will be in upward direction.

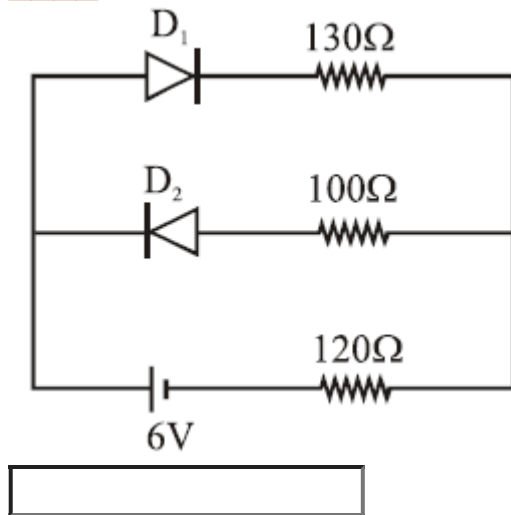


$$N' = M (g - a)$$

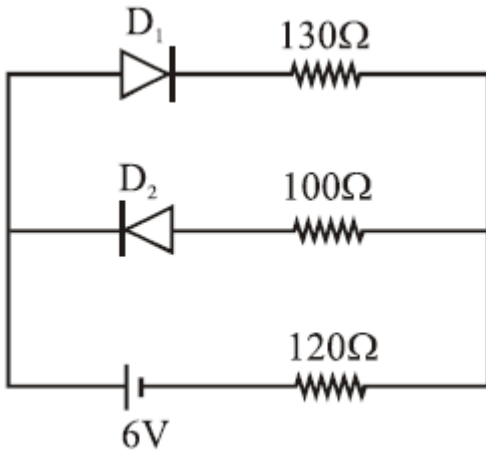
$$\Rightarrow 60 (10 - 1.8)$$

$$N' \Rightarrow 492 N$$

Q29: The circuit contains two diodes each with a forward resistance of $50\ \Omega$ and with infinite reverse resistance. If the battery voltage is 6 V , the current through the $120\ \Omega$ resistance is _____ mA.

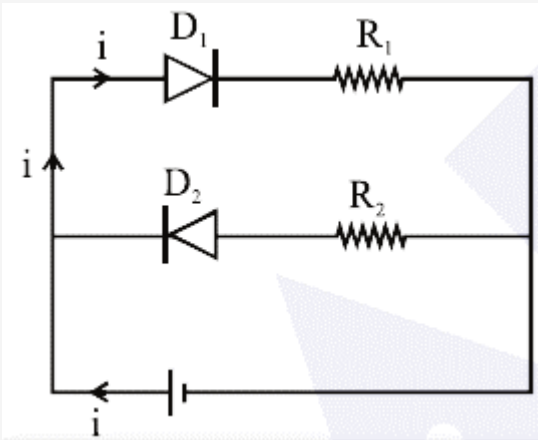


Q29: The circuit contains two diodes each with a forward resistance of $50\ \Omega$ and with infinite reverse resistance. If the battery voltage is 6 V , the current through the $120\ \Omega$ resistance is _____ mA.



20.00

Solution:



In this circuit D_1 will be forward bias and D_2 will be reverse bias.

$\therefore$ There will be no current through D_2 and R_2

Apply KVL in circuit we get

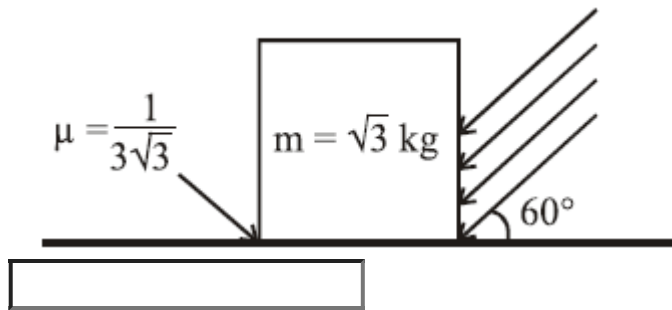
$$+6 - 50i - 130i - 120i = 0$$

$$i = \frac{6}{300} A = \frac{6}{300} \times 1000 mA$$

$$\Rightarrow 20 mA$$

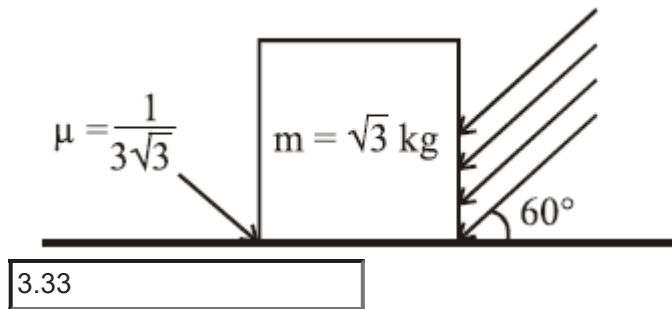
Q30: As shown in the figure, a block of mass $\sqrt{3}\text{kg}$ is kept on a horizontal rough surface of coefficient of friction $\frac{1}{3\sqrt{3}}$. The critical force to be applied on the vertical surface as shown at an angle 60° with horizontal such that it does not move, will be $3x$. The value of x will be _____.

$[g = 10 \text{ m/s}^2 ; \sin 60^\circ = \frac{\sqrt{3}}{2} ; \cos 60^\circ = \frac{1}{2}]$

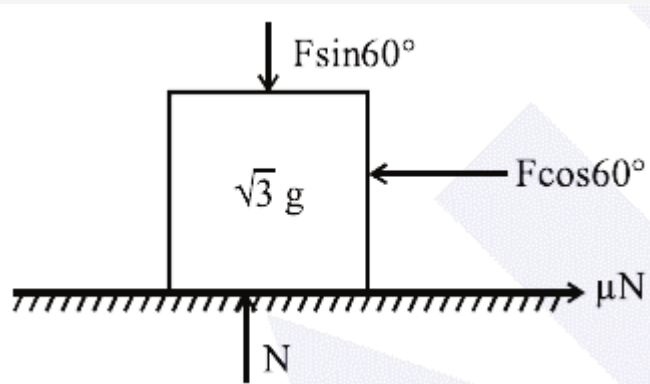


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$[g = 10 \text{ m/s}^2 ; \sin 60^\circ = \frac{\sqrt{3}}{2} ; \cos 60^\circ = \frac{1}{2}]$



Solution:



$$F \cos 60^\circ = \mu N \text{ or } \frac{F}{2} = \frac{1}{3\sqrt{3}} N \dots (1)$$

$$\& N = \sin 60^\circ + \sqrt{3}g \dots (2)$$

From equation (1) & (2)

$$\frac{F}{2} = \frac{1}{3\sqrt{3}} \left(\frac{F\sqrt{3}}{2} + \sqrt{3}g \right)$$

$$\Rightarrow F = g = 10 \text{ Newton} = 3x$$

$$\text{So, } x = \frac{10}{3} = 3.33$$

Q31: Match-I with List-II.

List-1	List-2
Electronic configuration of elements	$\Delta_f H$ in kJ mol^{-1}
(a) $1s^2 2s^2$	(i) 801
(b) $1s^2 2s^2 2p^4$	(ii) 899
(c) $1s^2 2s^2 2p^3$	(iii) 1314
(d) $1s^2 2s^2 2p^1$	(iv) 1402

Choose the most appropriate answer from the options given below :

(A) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)

(B) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (i), (d) $\rightarrow$ (ii)

(C) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (i), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (iii)

(D) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (ii)

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(D) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (ii)

Solution:

Order of IE is $B < Be < O < N$

Q32: Compound A used as a strong oxidizing agent is amphoteric in nature. It is the part of lead storage batteries. Compound A is :

- (A) PbO_2
- (B) PbSO_4
- (C) Pb_3O_4
- (D) PbO

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(B) PbSO_4

(C) Pb_3O_4

(D) PbO

Solution:

$\overset{+4}{\text{Pb}}\text{O}_2 \Rightarrow$ Amphoteric & oxidising agent.
 $\Rightarrow$ Used in lead storage battery.

Q33: Match List-1 with List-2.

List-1 (ore)	List-2 (Element Present)
(a) Kernite	(i) Tin
(b) Cassiterite	(ii) Boron
(c) Calamine	(iii) Fluorine
(d) Cryolite	(iv) Zinc

Choose the most appropriate answer from the options given below :

- (A) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (i), (d) $\rightarrow$ (iii)
- (B) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (iii)
- (C) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (iv)
- (D) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (ii)

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(A) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (i), (d) $\rightarrow$ (iii)

(B) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (iii)

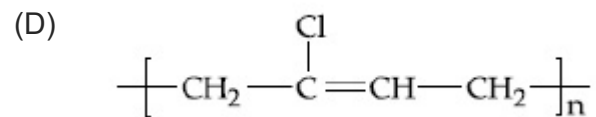
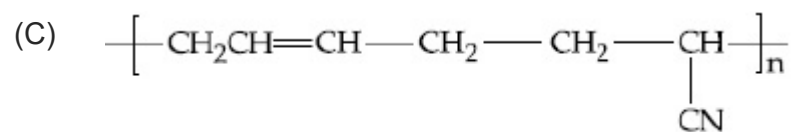
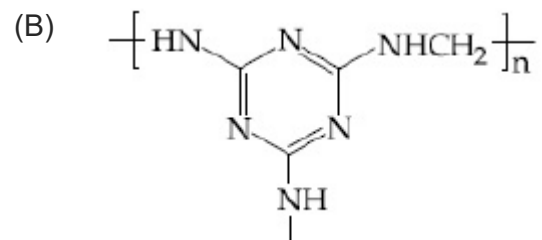
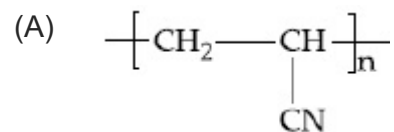
(C) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (iv)

(D) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (ii)

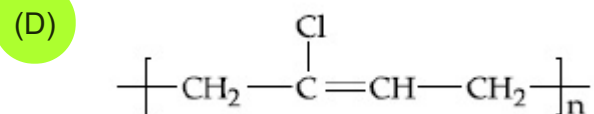
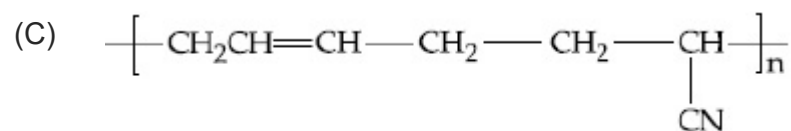
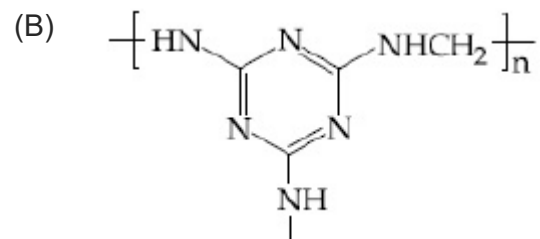
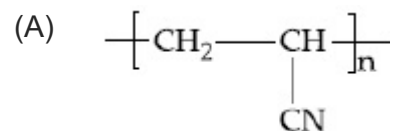
Solution:

	Ore	Formula
(a)	Cryolite	- Na_3AlF_6
(b)	Calamine	- $ZnCO_3$
(c)	Cassiterite	- SnO_2
(d)	Kernite	- $Na_2B_4O_7 \cdot 4H_2O$

Q34: The structure of Neoprene is:



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Solution:

Neoprene is polymer of Chloroprene $\left(\text{CH}_2 = \underset{\text{Cl}}{\text{C}} - \text{CH} = \text{CH}_2 \right)$

Q35: Which one of the following lanthanoids does not form MO_2 ? [M is lanthanoid metal]

(A) Dy

(B) Nd

(C) Yb

(D) Pr

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(B) Nd

(C) Yb

(D) Pr

Solution:

Nd, Dy and Pr exhibit +4 state in oxide MO_2 .

Q36: The orbital having two radial as well as two angular nodes is :

- (A) 3p
- (B) 5d
- (C) 4f
- (D) 4d

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(A) 3p

(B) 5d

(C) 4f

(D) 4d

Solution:

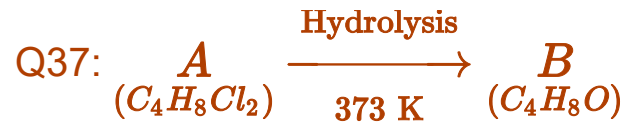
$$\text{Angular node} = \ell = 2$$

$$\text{Radial node} = (n - \ell - 1) = 2$$

$$= n - 2 - 1 = 2$$

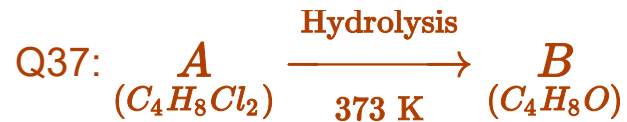
$$n = 5$$

So orbital is 5d



B reacts with Hydroxyl amine but does not give Tollen's test. Identify A and B.

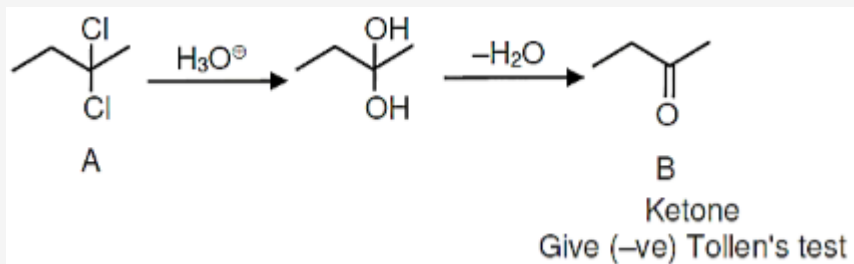
- (A) 2,2-Dichlorobutane and Butanal
- (B) 1,1-Dichlorobutane and 2-Butanone
- (C) 2,2-Dichlorobutane and Butan-2-one
- (D) 1, 1-Dichlorobutane and Bu tan al



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Solution:



Q38: The presence of ozone in troposphere

- (A) protects us from the radiation
- (B) protects us from greenhouse effect
- (C) protects us from the UV radiation
- (D) generates photochemical smog

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Solution:

The presence of ozone in the troposphere generates photochemical smog.

Q39: On treating a compound with warm dil. H_2SO_4 , gas X is evolved which turns $\text{K}_2\text{Cr}_2\text{O}_7$ solution acidified with dil. H_2SO_4 to a green compound Y.

X and Y respectively are :

- (A) $\text{X} = \text{SO}_2$, $\text{Y} = \text{Cr}_2(\text{SO}_4)_3$
- (B) $\text{X} = \text{SO}_2$, $\text{Y} = \text{Cr}_2\text{O}_3$
- (C) $\text{X} = \text{SO}_3$, $\text{Y} = \text{Cr}_2(\text{SO}_4)_3$
- (D) $\text{X} = \text{SO}_3$, $\text{Y} = \text{Cr}_2\text{O}_3$

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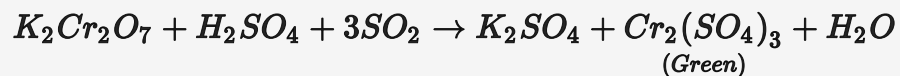
(A) X = SO_2 , Y = $Cr_2(SO_4)_3$

(B) X = SO_2 , Y = Cr_2O_3

(C) X = SO_3 , Y = $Cr_2(SO_4)_3$

(D) X = SO_3 , Y = Cr_2O_3

Solution:



Q40: Which of the following is 'a' FALSE statement?

- (A) Kjeldahl's method is used for the estimation of nitrogen in an organic compound
- (B) Carius tube is used in the estimation of sulphur in an organic compound
- (C) Carius method is used for the estimation of nitrogen in an organic compound
- (D) Phosphoric acid produced on oxidation of phosphorus present in an organic compound is precipitated as $\text{Mg}_2\text{P}_2\text{O}_7$ by adding magnesia mixture

Q40: Which of the following is 'a' FALSE statement?

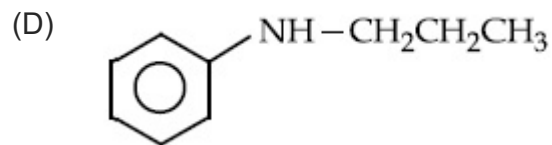
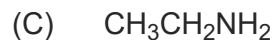
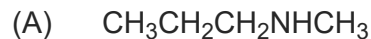
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Solution:

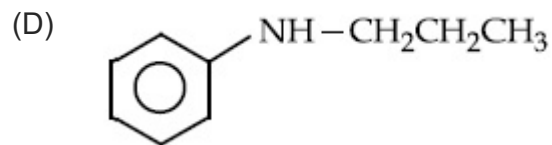
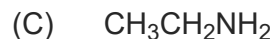
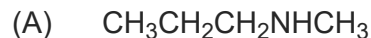
Carius method is used for estimation of S & halogens.

Kjeldhal's and Dumas method is used for nitrogen estimation.

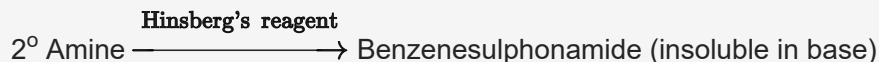
Q41: An amine on reaction with benzene sulphonyl chloride produces a compound insoluble in alkaline solution. This amine can be prepared by ammonolysis of ethyl chloride. The correct structure of amine is :



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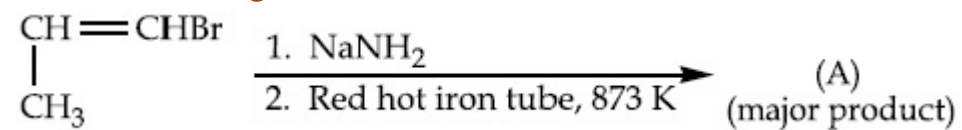


Solution:

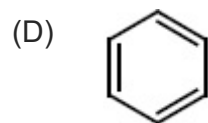
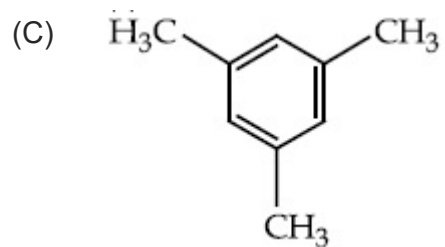
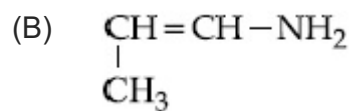
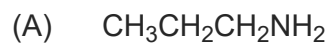


Amine which is formed by ammonolysis of ethyl chloride should contain atleast one ethyl group.

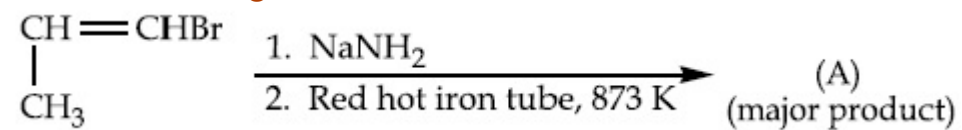
Q42: For the given reaction:



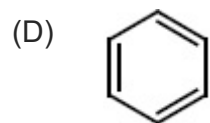
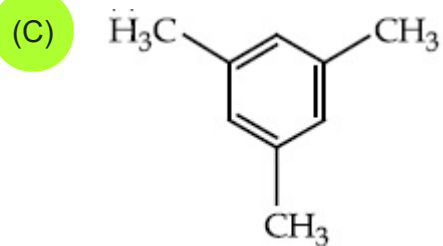
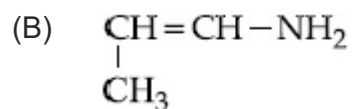
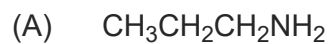
What is 'A'?



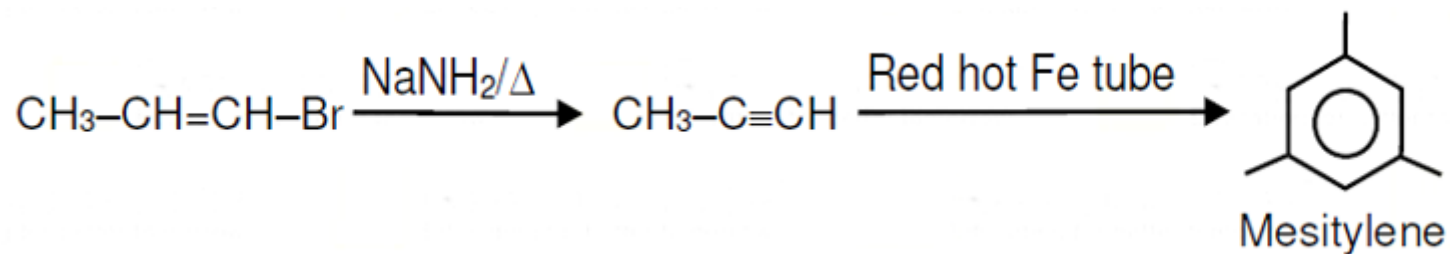
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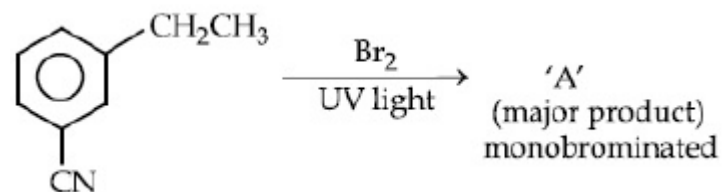
What is 'A'?



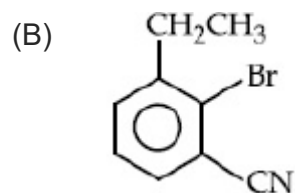
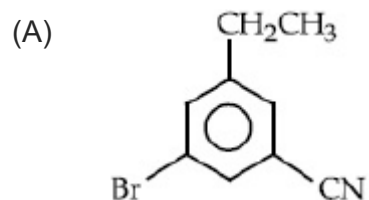
Solution:



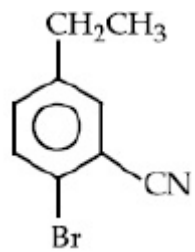
Q43: For the given reaction :



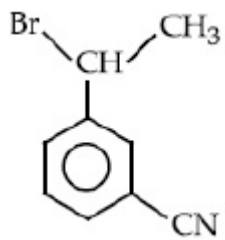
What is 'A'?



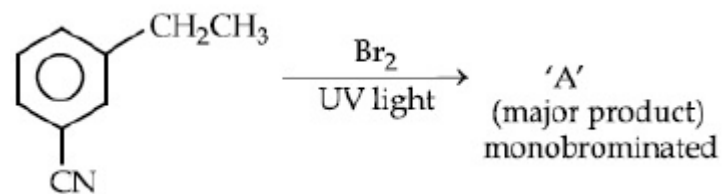
(C)



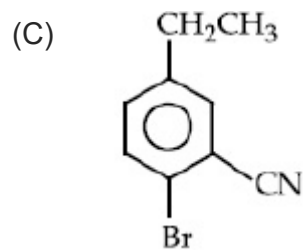
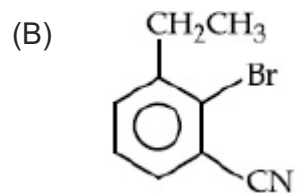
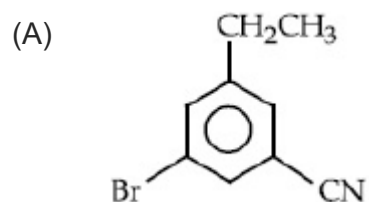
(D)



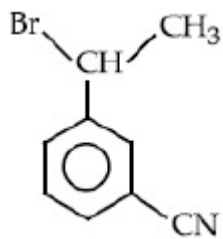
Q43: For the given reaction :



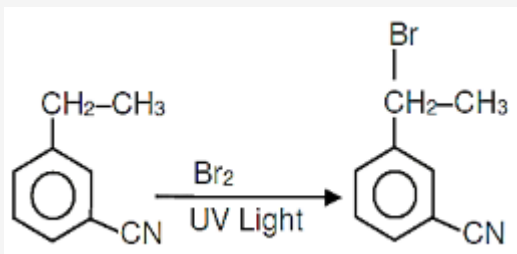
What is 'A'?



(D)

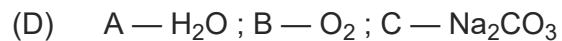
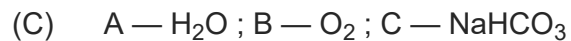
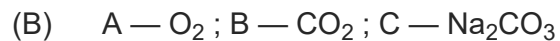


Solution:

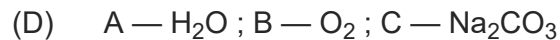
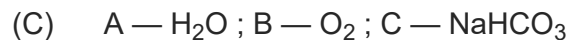
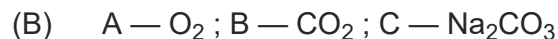


Based on free radical stability by free radical substitution mechanism.

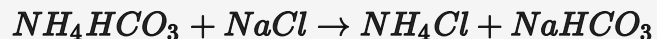
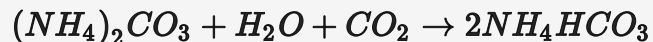
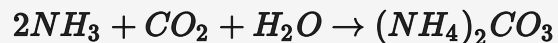
Q44: Find A, B and C in the following reactions :



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Solution:



Q45: Given below are two statements :

Statement I : A mixture of chloroform and aniline can be separated by simple distillation.

Statement II : When separating aniline from a mixture of aniline and water by steam distillation aniline boils below its boiling point.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Both statement I and statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is false but Statement II is true
- (D) Statement I is true but Statement II is false

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- (D) Statement I is true but Statement II is false

Solution:

Statement 1 : B.P. of chloroform = 334 K

B.P. of aniline = 457 K

thus can be separated of simple distillation.

Statement 2 : Mixture of aniline and water separated by simple distillation.

Q46: Given below are two statements : one is labelled as Assertion /1. and the other is labelled as Reason R.

Assertion R : Dipole-dipole interactions are the only non-covalent interactions, resulting in hydrogen bond formation.

Reason R : Fluorine is the most electronegative element and hydrogen bonds in HF are symmetrical.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) A is true but R is false
- (B) Both A and R are true but R is NOT the correct explanation of A
- (C) Both A and R are true and R is the correct explanation of A
- (D) A is false but R is true

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- (C) Both A and R are true and R is the correct explanation of A
- (D) A is false but R is true

Solution:

Hydrogen bond is attractive force which binds H atom of one molecule with electronegative atom [F, O,N] of another molecule, It is electro-static attraction between the pole of H atom and lone pair of electron on other electro negative atom.

Q47: Given below are two statements :

Statement I : o-Nitrophenol is steam volatile due to intramolecular hydrogen bonding.

Statement II : o-Nitrophenol has high melting due to hydrogen bonding.

In the light of the above statements, choose the most appropriate answer from the options given below

- (A) Both statement I and statement II are false
- (B) statement I is false but statement II is true
- (C) Statement I is true but statement II is false
- (D) Both statement I and statement II are true

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- (A) Both statement I and statement II are false
- (B) statement I is false but statement II is true
- ☒ (C) Statement I is true but statement II is false
- (D) Both statement I and statement II are true

Solution:

Ortho nitro phenol is steam volatile due to Intramolecular H-bonding while p-nitro phenol is less volatile due to intermolecular H-bonding.

p-nitro phenol has high melting point due to more symmetric in nature.

Q48: Which of the following vitamin is helpful in delaying the blood clotting ?

- (A) Vitamin E
- (B) Vitamin C
- (C) Vitamin K
- (D) Vitamin B

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- (A) Vitamin E
- (B) Vitamin C
- ☒ (C) Vitamin K
- (D) Vitamin B

Solution:

Vitamin K is used by the body to help blood clot.

Q49: Statements about heavy water are given below.

A. Heavy water is used in exchange reactions for the study of reaction mechanisms.

B. Heavy water is prepared by exhaustive electrolysis of water.

C. Heavy water has higher boiling point than ordinary water.

D. Viscosity of H_2O is greater than D_2O .

Choose the most appropriate answer from the options given below :

(A) A, B and C only

(B) A and D only

(C) A and C only

(D) A and B only

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Choose the most appropriate answer from the options given below :

(A) A, B and C only

(B) A and D only

(C) A and C only

(D) A and B only

Solution:

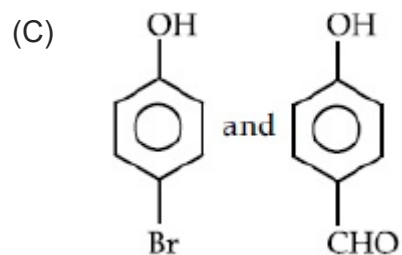
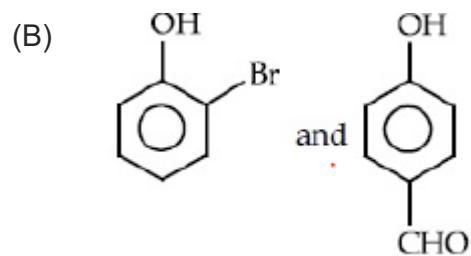
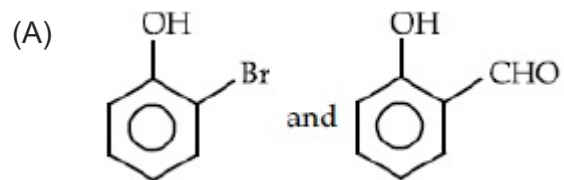
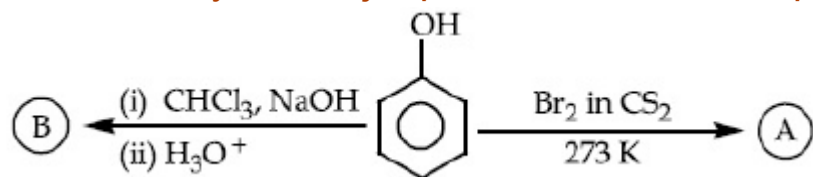
(A) Chemical reactions of heavy water are slower than water.

(B) Heavy water prepared by exhaustive electrolysis of water.

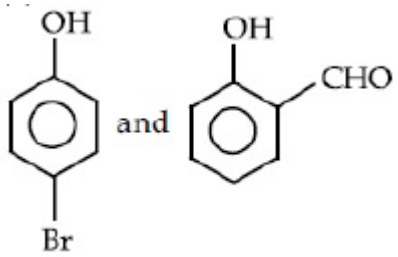
(C) $(BP)_{D_2O} > (BP)_{H_2O}$

(D) Viscosity of $D_2O >$ Viscosity of H_2O

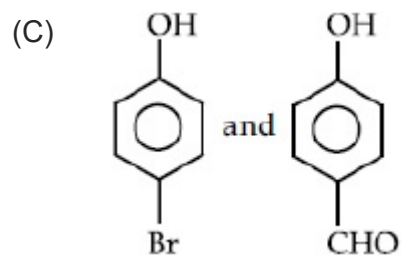
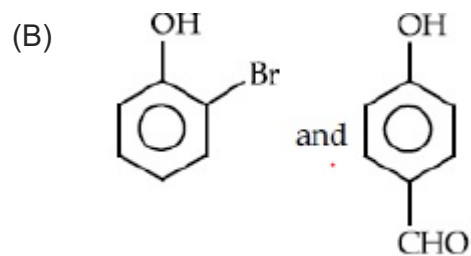
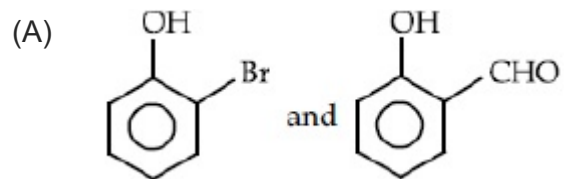
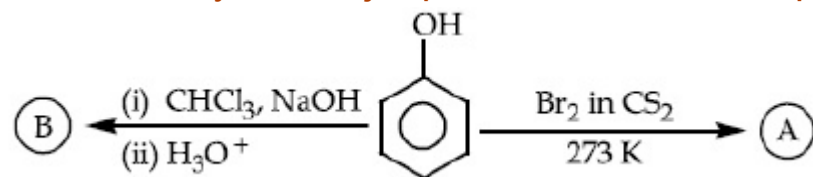
Q50: Identify the major products A and B respectively in the following reactions of phenol :



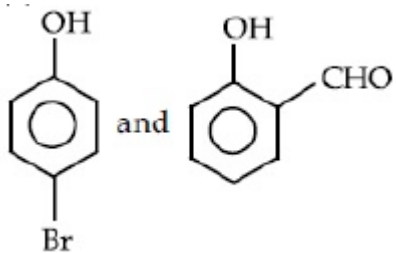
(D)



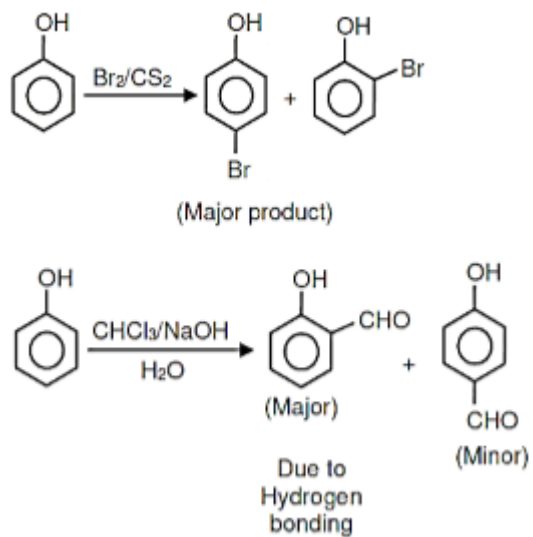
Q50: Identify the major products A and B respectively in the following reactions of phenol :



(D)



Solution:



(Reimer Tiemann reaction)

Q51: Consider the following reaction



The quantity of electricity required in Faraday to reduce five moles of MnO_4^- is ____.

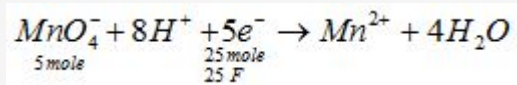
Q51: Consider the following reaction



The quantity of electricity required in Faraday to reduce five moles of MnO_4^- is ____.

25.00

Solution:



Q52: For a chemical reaction $A + B \rightleftharpoons C + D$

$(\Delta_r H^\ominus = 80 \text{ kJ mol}^{-1})$ the entropy change $\Delta_r S^\ominus$ depends on the temperature T (in K) as $\Delta_r S^\ominus = 2T \text{ (J K}^{-1}\text{mol}^{-1}\text{)}$.

Minimum temperature at which it will become spontaneous is _____ K. (Integer)

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Minimum temperature at which it will become spontaneous is _____K. (Integer)

200.00

Solution:

For a reaction to be spontaneous

$$\Delta G = (\Delta H - T\Delta S) < 0$$

As $\Delta H = +ve$ & $\Delta S = +ve$

So reaction will be spontaneous above equilibrium temperature.

At equilibrium $\Delta H - T\Delta S = 0$

$$80 - \frac{T[2T]}{1000} = 0$$

$$80 = \frac{2T^2}{1000}$$

$$T^2 = 40 \times 10^3$$

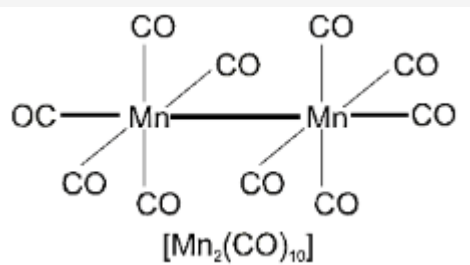
$$T = 200K$$

Q53: Number of bridging CO ligands in $[\text{Mn}_2(\text{CO})_{10}]$ is

Q53: Number of bridging CO ligands in $[\text{Mn}_2(\text{CO})_{10}]$ is

0.00

Solution:



So there is no bridging CO ligand

Q54: An exothermic reaction $X \rightarrow Y$ has an activation energy 30 kJ mol^{-1} . If energy change ΔE during the reaction is -20 kJ , then the activation energy for the reverse reaction in kJ is

 .

Q54: An exothermic reaction $X \rightarrow Y$ has an activation energy 30 kJ mol^{-1} . If energy change ΔE during the reaction is -20 kJ , then the activation energy for the reverse reaction in kJ is

50.00

Solution:

$$\Delta H_{\text{reaction}} = (E_{af}) - (E_{ab})$$

$$-20 = 30 - (E_{ab})$$

$$(E_{ab}) = 50 \text{ kJ/mole}$$

Q55: A certain gas obeys $P(V_m - b) = RT$. The value of $\left(\frac{\partial Z}{\partial P}\right)_T$ is $\frac{xb}{RT}$. The value of x is _____. (Z : compressibility factor)

Q55: A certain gas obeys $P(V_m - b) = RT$. The value of $\left(\frac{\partial Z}{\partial P}\right)_T$ is $\frac{xb}{RT}$. The value of x is _____. (Z : compressibility factor)

1.00

Solution:

$$P(V_m - b) = RT$$

$$PV_m - Pb = RT$$

$$\left(\frac{PV_m}{RT}\right) - \frac{Pb}{RT} = 1$$

$$Z - \frac{Pb}{RT} = 1$$

$$\text{So } Z = 1 + \frac{Pb}{RT}$$

$$\left(\frac{\partial Z}{\partial P}\right)_T = \frac{b}{RT}$$

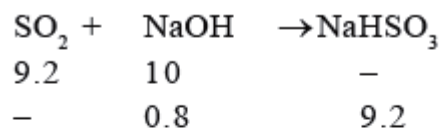
$$\text{So } x = 1$$

Q56: 224 mL of $\text{SO}_{2(g)}$ at 298 K and 1 atm is passed through 100 mL of 0.1 M NaOH solution. The non-volatile solute produced is dissolved in 36 g of water. The lowering of vapour pressure of solution (assuming the solution is dilute) ($P_{\text{H}_2\text{O}}^{\circ} = 24 \text{ mm of Hg}$) is $x \times 10^{-2} \text{ mm of Hg}$, the value of x is _____. (Integer answer)

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24.00

Solution:



$$\Delta P = P^{\circ} \cdot X_{\text{solute}}$$

$$= 24 \times \frac{(1.6 + 18.4)}{2020}$$

$$= 0.2376 = 23.76 \times 10^{-2}$$

$$= 24 \times 10^{-2}$$

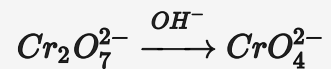
Q57: Dichromate ion is treated with base, the oxidation number of Cr in the product formed is

Q57: Dichromate ion is treated with base, the oxidation number of Cr in the product formed is

6.00

Solution:

In basic medium dichromate ion convert to chromate ion



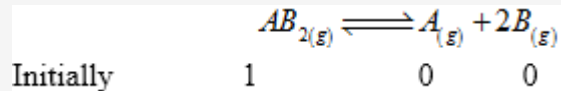
Here O.S of Cr in CrO_4^{2-} is +6

Q58: A homogeneous ideal gaseous reaction $AB_{2(g)} = A_{(g)} + 2B_{(g)}$ is carried out in a 25 litre flask at 27°C . The initial amount of AB_2 was 1 mole and the equilibrium pressure was 1.9 atm. The value of K_p is $x \times 10^{-2}$. The value of x is _____. (Integer answer)

Q58: A homogeneous ideal gaseous reaction $AB_{2(g)} = A_{(g)} + 2B_{(g)}$ is carried out in a 25 litre flask at 27°C. The initial amount of AB_2 was 1 mole and the equilibrium pressure was 1.9 atm. The value of K_p is $x \times 10^{-2}$. The value of x is _____. (Integer answer)

73.00

Solution:



At Equilibrium $(1 - \alpha) \alpha \ 2\alpha$

$$(n_{total})_{at \ equilibrium} = (1 + 2\alpha)$$

At Equilibrium,

$$PV = n_{total}RT$$

$$1.9 \times 25 = n_{total} \times 0.082 \times 300$$

$$n_{total} = 1.93$$

$$n_{total} = (1 + 2\alpha) = 1.93$$

$$\alpha = 0.465$$

$$K_p = \frac{(p_A)(p_B)^2}{(p_{AB_2})} = \frac{\left[\frac{0.465}{1.93} \times 1.9\right] \left[\frac{0.93}{1.93} \times 1.9\right]^2}{\left[\frac{0.535}{1.93} \times 1.9\right]} = 0.727 \approx 72.7 \times 10^{-2}$$

$$So \ x = 72.7 \approx 73$$

Q59: 3.12 g of oxygen is adsorbed on 1.2 g of platinum metal. The volume of oxygen adsorbed per gram of the adsorbent at 1 atm and 300 K in L is _____.

Q59: 3.12 g of oxygen is adsorbed on 1.2 g of platinum metal. The volume of oxygen adsorbed per gram of the adsorbent at 1 atm and 300 K in L is _____.

2.00

Solution:

$$PV = nRT$$

$$1 \times V = \left[\frac{3.12}{32} \right] \times 0.0821 \times 300$$

$$V = 2.4 \text{ lit.}$$

$$\text{Volume of } O_2 \text{ absorbed by per gram of adsorbent} = \frac{2.4}{1.2} = 2$$

Q60: The number of significant figures in 50000.020×10^{-3} is

Q60: The number of significant figures in 50000.020×10^{-3} is

Solution:

follow the rules of significant figures

Q61: The number of seven digit integers with sum of the digits equal to 10 and formed by using the digits 1, 2 and 3 only is :

(A) 35

(B) 82

(C) 77

(D) 42

Q61: The number of seven digit integers with sum of the digits equal to 10 and formed by using the digits 1, 2 and 3 only is :

(A) 35

(B) 82

(C) 77

(D) 42

Solution:

Case 1: 1, 1, 1, 1, 1, 2, 3

$$\text{ways} = \frac{7!}{5!} = 42$$

Case 2: 1, 1, 1, 1, 2, 2, 2

$$\text{ways} = \frac{7!}{4!.3!} = 35$$

$$\text{Total ways} = 42 + 35 = 77$$

Q62: A fair coin is tossed a fixed number of times. If the probability of getting 7 heads is equal to probability of getting 9 heads, then the probability of getting 2 heads is :

(A) $\frac{15}{2^{13}}$

(B) $\frac{15}{2^8}$

(C) $\frac{15}{2^{14}}$

(D) $\frac{15}{2^{12}}$

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(B) $\frac{15}{2^8}$

(C) $\frac{15}{2^{14}}$

(D) $\frac{15}{2^{12}}$

Solution:

$${}^nC_9 \times \left(\frac{1}{2}\right)^9 \times \left(\frac{1}{2}\right)^{n-9} = {}^nC_7 \times \left(\frac{1}{2}\right)^7 \times \left(\frac{1}{2}\right)^{n-7}$$

$${}^nC_9 = {}^nC_7 \Rightarrow n = 16$$

$$P(2 \text{ Heads}) = {}^{16}C_2 \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^{14}$$

$$= {}^{16}C_2 \times \left(\frac{1}{2}\right)^{16}$$

$$= \frac{15}{2^{13}}$$

Q63: Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is :

(A) 12

(B) 4

(C) 6

(D) 1

Q63: Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is :

(A) 12

(B) 4

(C) 6

(D) 1

Solution:

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

$$A^2 = \begin{bmatrix} a^2 + b^2 & b(a+c) \\ b(a+c) & b^2 + c^2 \end{bmatrix}$$

$$\text{tr}(A^2) = a^2 + 2b^2 + c^2 = 1$$

$$\Rightarrow b = 0 \text{ and } a^2 + c^2 = 1$$

$$\Rightarrow (a, c) \equiv (1, 0), (-1, 0), (0, 1), (0, -1)$$

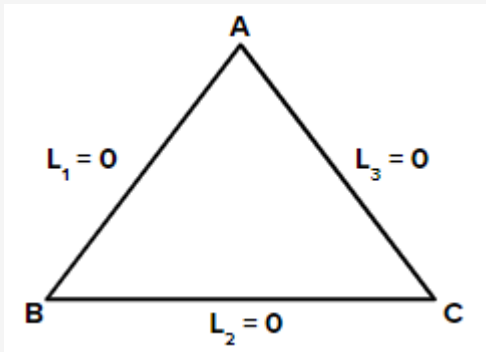
Q64: The intersection of three lines $x - y = 0$, $x + 2y = 3$ and $2x + y = 6$ is a :

- (A) Equilateral triangle
- (B) Right angled triangle
- (C) Isosceles triangle
- (D) None of the above

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- (A) Equilateral triangle
- (B) Right angled triangle
- (C) Isosceles triangle
- (D) None of the above

Solution:



$$L_1 : x - y = 0$$

$$L_2 : x + 2y = 3$$

$$L_3 : 2x + y = 6$$

$$A(2, 2)$$

$$B(1, 1)$$

$$C(3, 0)$$

$$\Rightarrow AB = \sqrt{2}, BC = \sqrt{5}, AC = \sqrt{5}$$

$\therefore$ Triangle is isosceles

Q65: Let f be any function defined on $\mathbb{R}$ and let it satisfy the condition

$$|f(x) - f(y)| \leq |x - y|^2, \forall (x, y) \in \mathbb{R}$$

If $f(0) = 1$, then :

- (A) $f(x) > 0, \forall x \in \mathbb{R}$
- (B) $f(x) = 0, \forall x \in \mathbb{R}$
- (C) $f(x) < 0, \forall x \in \mathbb{R}$
- (D) $f(x)$ can take any value in $\mathbb{R}$

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(D) $f(x)$ can take any value in $\mathbb{R}$

Solution:

$$\left| \frac{f(x) - f(y)}{x - y} \right| \leq |x - y|$$

$$\Rightarrow |f'(x)| \leq 0$$

$$\Rightarrow f'(x) = 0$$

$$\Rightarrow f(x) = \text{constant}$$

$$\Rightarrow f(x) = 1$$

Q66: The value of $\lim_{h \rightarrow 0} 2 \left\{ \frac{\sqrt{3} \sin\left(\frac{\pi}{6} + h\right) - \cos\left(\frac{\pi}{6} + h\right)}{\sqrt{3}h(\sqrt{3} \cos h - \sin h)} \right\}$ is :

(A) $\frac{2}{\sqrt{3}}$

(B) $\frac{2}{3}$

(C) $\frac{4}{3}$

(D) $\frac{3}{4}$

Q66: The value of $\lim_{h \rightarrow 0} 2 \left\{ \frac{\sqrt{3} \sin\left(\frac{\pi}{6} + h\right) - \cos\left(\frac{\pi}{6} + h\right)}{\sqrt{3}h(\sqrt{3} \cos h - \sin h)} \right\}$ is :

(A) $\frac{2}{\sqrt{3}}$

(B) $\frac{2}{3}$

(C) $\frac{4}{3}$

(D) $\frac{3}{4}$

Solution:

$$\begin{aligned} & \lim_{h \rightarrow 0} 2 \cdot \frac{\left(\frac{\sqrt{3}}{2} \sin\left(h + \frac{\pi}{6}\right) - \frac{1}{2} \cos\left(h + \frac{\pi}{6}\right)\right)}{\sqrt{3}h\left(\frac{\sqrt{3}}{2} \cosh - \frac{1}{2} \sinh\right)} \\ & \lim_{h \rightarrow 0} 2 \cdot \frac{\left(\cos \frac{\pi}{6} \sin\left(h + \frac{\pi}{6}\right) - \sin \frac{\pi}{6} \cos\left(h + \frac{\pi}{6}\right)\right)}{\sqrt{3}h\left(\sin \frac{\pi}{3} \cos h - \cos \frac{\pi}{3} \sin h\right)} \\ & \lim_{h \rightarrow 0} 2 \cdot \frac{\sin\left(h + \frac{\pi}{6} - \frac{\pi}{6}\right)}{\sqrt{3}h \cdot \sin\left(\frac{\pi}{3} - h\right)} \\ & \lim_{h \rightarrow 0} \frac{2 \cdot 1}{\sqrt{3} \sin \frac{\pi}{3}} = \frac{2}{3} \cdot 2 = \frac{4}{3} \end{aligned}$$

Q67: If $(1, 5, 35)$, $(7, 5, 5)$, $(1, \lambda, 7)$ and $(2\lambda, 1, 2)$ are coplanar, then the sum of all possible values of λ is :

(A) $\frac{44}{5}$

(B) $\frac{39}{5}$

(C) $-\frac{44}{5}$

(D) $-\frac{39}{5}$

Q67: If $(1, 5, 35)$, $(7, 5, 5)$, $(1, \lambda, 7)$ and $(2\lambda, 1, 2)$ are coplanar, then the sum of all possible values of λ is :

(A) $\frac{44}{5}$

(B) $\frac{39}{5}$

(C) $-\frac{44}{5}$

(D) $-\frac{39}{5}$

Solution:

If points are coplanar then

$$\begin{vmatrix} 6 & 0 & -30 \\ 0 & \lambda - 5 & -28 \\ 2\lambda - 1 & -4 & -33 \end{vmatrix} = 0$$

$$6[-33\lambda + 165 - 112] + 30[2\lambda^2 - 10\lambda - \lambda + 5] = 0$$

$$-198\lambda + 318 + 60\lambda^2 - 330\lambda + 150 = 0$$

$$60\lambda^2 - 528\lambda + 468 = 0$$

$$\text{Sum of all value of } \lambda = \frac{528}{60} = \frac{44}{5}$$

Q68: If $\frac{\sin^{-1}x}{a} = \frac{\cos^{-1}x}{b} = \frac{\tan^{-1}y}{c}$; $0 < x < 1$, then the value of $\cos\left(\frac{\pi c}{a+b}\right)$ is :

(A) $\frac{1-y^2}{1+y^2}$

(B) $\frac{1-y^2}{y\sqrt{y}}$

(C) $\frac{1-y^2}{2y}$

(D) $1 - y^2$

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(C) $\frac{1-y^2}{2y}$

(D) $1 - y^2$

Solution:

$$\text{Let } \sin^{-1}x = a\lambda, \cos^{-1}x = b\lambda, \tan^{-1}y = c\lambda$$

$$\Rightarrow (a+b)\lambda = \frac{\pi}{2}$$

$$\Rightarrow \frac{\pi}{a+b} = 2\lambda$$

$$\text{Now, } \cos\left(\frac{\pi}{a+b}\right) = \cos(2\lambda c) = \cos(2\tan^{-1}y) = \frac{1-y^2}{1+y^2}$$

Q69: In an increasing geometric series, the sum of the second and the sixth term is $\frac{25}{2}$ and the product of the third and fifth term is 25. Then, the sum of 4^{th} , 6^{th} and 8^{th} terms is equal to :

- (A) 26
- (B) 30
- (C) 32
- (D) 35

Q69: In an increasing geometric series, the sum of the second and the sixth term is $\frac{25}{2}$ and the product of the third and fifth term is 25. Then, the sum of 4^{th} , 6^{th} and 8^{th} terms is equal to :

(A) 26

(B) 30

(C) 32

(D) 35

Solution:

$$ar + ar^4 = \frac{25}{2} \text{ and } ar^5 \cdot ar^4 = 25 \Rightarrow ar^3 = 5$$

$$\therefore \frac{r+r^2}{r^3} = \frac{5}{2}$$

$$\Rightarrow 2 + 2r^4 = 5r^2$$

$$\Rightarrow 2r^4 - 5r^2 + 2 = 0$$

$$\Rightarrow r^2 = 2 \text{ or } r^2 = \frac{1}{2} \text{ Reject}$$

$$\text{Now, } ar^3 + ar^5 + ar^7 = 5 + ar^5(1 + r^2) = 5 + 5.2(1 + 2) = 35$$

Q70: The value of $\begin{vmatrix} (a+1)(a+2) & a+2 & 1 \\ (a+2)(a+3) & a+3 & 1 \\ (a+3)(a+4) & a+4 & 1 \end{vmatrix}$ is :

(A) $(a+2)(a+3)(a+4)$

(B) 0

(C) -2

(D) $(a+1)(a+2)(a+3)$

Q70: The value of $\begin{vmatrix} (a+1)(a+2) & a+2 & 1 \\ (a+2)(a+3) & a+3 & 1 \\ (a+3)(a+4) & a+4 & 1 \end{vmatrix}$ is :

(A) $(a+2)(a+3)(a+4)$

(B) 0

(C) -2

(D) $(a+1)(a+2)(a+3)$

Solution:

$$\begin{aligned} R_3 &\rightarrow R_3 - R_2, R_2 \rightarrow R_2 - R_1 \\ &= \begin{vmatrix} (a+1)(a+2) & a+2 & 1 \\ 2(a+2) & 1 & 0 \\ 2(a+3) & 1 & 0 \end{vmatrix} \\ &= 2a+4-2a-6 = -2 \end{aligned}$$

Q71: The maximum slope of the curve $y = \frac{1}{2}x^4 - 5x^3 + 18x^2 - 19x$ occurs at the point:

(A) $(3, \frac{21}{2})$

(B) $(0, 0)$

(C) $(2, 9)$

(D) $(2, 2)$

Q71: The maximum slope of the curve $y = \frac{1}{2}x^4 - 5x^3 + 18x^2 - 19x$ occurs at the point:

(A) $(3, \frac{21}{2})$

(B) $(0, 0)$

(C) $(2, 9)$

(D) $(2, 2)$

Solution:

$$y = \frac{1}{2}x^4 - 5x^3 + 18x^2 - 19x$$

$$\frac{dy}{dx} = \text{slope}$$

$$\therefore \frac{dy}{dx} = 2x^3 - 15x^2 + 36x - 19$$

Now the maximum and minimum of slope we will differentiate slope

$$\therefore \frac{d^2y}{dx^2} = 6x^2 - 30x + 36$$

$$= 6(x^2 - 5x + 6)$$

$$= 6(x - 2)(x - 3)$$

$$\frac{d^2y}{dx^3} = 12x - 30$$

$$\text{at } x = 2 \quad \frac{d^2y}{dx^2} < 0$$

Hence $x = 2$ point of maximum

$\therefore$ Point is $(2, 2)$

Q72: Let $R = \{(P, Q) \mid P \text{ and } Q \text{ are at the same distance from the origin}\}$ be a relation, then the equivalence class of $(1, -1)$ is the set :

(A) $S = \{(x, y) \mid x^2 + y^2 = 4\}$

(B) $S = \{(x, y) \mid x^2 + y^2 = 1\}$

(C) $S = \{(x, y) \mid x^2 + y^2 = 2\}$

(D) $S = \{(x, y) \mid x^2 + y^2 = \sqrt{2}\}$

Q72: Let $R = \{(P, Q) \mid P \text{ and } Q \text{ are at the same distance from the origin}\}$ be a relation, then the equivalence class of $(1, -1)$ is the set :

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(C) $S = \{(x, y) \mid x^2 + y^2 = 2\}$

(D) $S = \{(x, y) \mid x^2 + y^2 = \sqrt{2}\}$

Solution:

As the distance from origin of the point $(1, -1)$ is $\sqrt{2}$ so the locus of all the points at $\sqrt{2}$ distance is $x^2 + y^2 = 2$.
Hence the equivalence class containing $(1, -1)$ for the relation $x^2 + y^2 = 2$

Q73: The maximum value of the term independent of 't' in the expansion of

$$\left(tx^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t} \right)^{10} \text{ where } x \in (0, 1) \text{ is}$$

(A) $\frac{10!}{3(5!)^2}$

(B) $\frac{2 \cdot 10!}{3\sqrt{3}(5!)^2}$

(C) $\frac{10!}{\sqrt{3}(5!)^2}$

(D) $\frac{2 \cdot 10!}{3(5!)^2}$

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(B) $\frac{2 \cdot 10!}{3\sqrt{3}(5!)^2}$

(C) $\frac{10!}{\sqrt{3}(5!)^2}$

(D) $\frac{2 \cdot 10!}{3(5!)^2}$

Solution:

$$T_{r+1} = {}^{10}C_r \left(tx^{\frac{1}{5}} \right)^{10-r} \left(\frac{(1-x)^{\frac{1}{10}}}{t} \right)^r$$

$$10 - r - r = 0 \Rightarrow r = 5$$

$$\frac{d(T_6)}{dx} = {}^{10}C_5 \left((1-x)^{1/2} + \frac{-x}{2\sqrt{1-x}} \right) = 0$$

$$2(1-x) - x = 0 \Rightarrow x = \frac{2}{3}$$

$$\text{Maximum } T_6 = {}^{10}C_5 \frac{2}{3} \left(\frac{1}{3} \right)^{\frac{1}{2}} = \frac{2 \cdot 10!}{3\sqrt{3}(5!)^2}$$

Q74: The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos^2 x}{1+3^x} dx$ is

(A) 2π

(B) $\frac{\pi}{2}$

(C) 4π

(D) $\frac{\pi}{4}$

Q74: The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos^2 x}{1+3^x} dx$ is

(A) 2π

(B) $\frac{\pi}{2}$

(C) 4π

(D) $\frac{\pi}{4}$

Solution:

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \left(\frac{\cos^2 x}{1+3^x} + \frac{\cos^2 x}{1+3^{-x}} \right) dx = \int_0^{\frac{\pi}{2}} \left(\frac{\cos^2 x}{1+3^x} + \frac{3^x \cos^2 x}{1+3^x} \right) dx = \int_0^{\frac{\pi}{2}} \cos^2 x dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 + \cos 2x) dx = \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) \Bigg|_0^{\frac{\pi}{2}} = \frac{1}{2} \left(\frac{\pi}{2} \right) = \frac{\pi}{4} \end{aligned}$$

Q75: If $\vec{a}$ and $\vec{b}$ are perpendicular, then $\vec{a} \times \left(\vec{a} \times \left(\vec{a} \times \left(\vec{a} \times \vec{b} \right) \right) \right)$ is equal to

(A) $\vec{0}$

(B) $\frac{1}{2}|\vec{a}|^4\vec{b}$

(C) $\vec{a} \times \vec{b}$

(D) $|\vec{a}|^4\vec{b}$

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(A) $\vec{0}$

(B) $\frac{1}{2} |\vec{a}|^4 \vec{b}$

(C) $\vec{a} \times \vec{b}$

(D) $|\vec{a}|^4 \vec{b}$

Solution:

$$\begin{aligned} & \vec{a} \times \left(\vec{a} \times \left(\left(\vec{a} \cdot \vec{b} \right) \vec{a} - |\vec{a}|^2 \vec{b} \right) \right) \\ &= \vec{a} \times \left(-|\vec{a}|^2 \left(\vec{a} \times \vec{b} \right) \right) = -|\vec{a}|^2 \left(\left(\vec{a} \cdot \vec{b} \right) \vec{a} - |\vec{a}|^2 \vec{b} \right) = |\vec{a}|^4 \vec{b} - |\vec{a}|^2 \left(\vec{a} \cdot \vec{b} \right) \vec{a} \\ &= |\vec{a}|^4 \vec{b} \left(\because \vec{a} \cdot \vec{b} = 0 \right) \end{aligned}$$

Q76: The Value of $\sum_{n=1}^{100} \int_{n-1}^n e^{x-[x]} dx$, where $[x]$ is the greatest integer $\leq x$, is

(A) $100(e - 1)$

(B) $100(1 + e)$

(C) $100(1 - e)$

(D) $100e$

Q76: The Value of $\sum_{n=1}^{100} \int_{n-1}^n e^{x-[x]} dx$, where $[x]$ is the greatest integer $\leq x$, is

(A) $100(e - 1)$

(B) $100(1 + e)$

(C) $100(1 - e)$

(D) $100e$

Solution:

$$\begin{aligned} & \sum_{n=1}^{100} \int_{n-1}^n e^{\{x\}} dx \\ &= 100 \int_0^1 e^x = 100(e - 1) \end{aligned}$$

Q77: The sum of the infinite series $1 + \frac{2}{3} + \frac{7}{3^2} + \frac{12}{3^3} + \frac{17}{3^4} + \frac{22}{3^5} + \dots$ is equal to

(A) $\frac{13}{4}$

(B) $\frac{15}{4}$

(C) $\frac{9}{4}$

(D) $\frac{11}{4}$

Q77: The sum of the infinite series $1 + \frac{2}{3} + \frac{7}{3^2} + \frac{12}{3^3} + \frac{17}{3^4} + \frac{22}{3^5} + \dots$ is equal to

(A) $\frac{13}{4}$

(B) $\frac{15}{4}$

(C) $\frac{9}{4}$

(D) $\frac{11}{4}$

Solution:

$$S = 1 + \frac{2}{3} + \frac{7}{3^2} + \frac{12}{3^3} + \dots \infty \quad \dots (i)$$

$$\frac{1}{3}S = \frac{1}{3} + \frac{2}{3^2} + \frac{7}{3^3} + \dots \quad \dots (ii)$$

(i) - (ii)

$$\frac{2}{3}S = 1 + \frac{1}{3} + \frac{5}{3^2} + \frac{5}{3^3} + \dots$$

$$\frac{2}{3}S = \frac{4}{3} + \frac{5}{3^2} + \frac{5}{3^3} + \dots$$

$$\frac{2}{3}S = \frac{4}{3} + \frac{\frac{5}{3^2}}{1 - \frac{1}{3}} = \frac{4}{3} + \frac{5}{6} = \frac{13}{6}$$

$$S = \frac{13}{6} \times \frac{3}{2} = \frac{13}{4}$$

Q78: Consider the three planes

$$P_1 : 3x + 15y + 21z = 9$$

$$P_2 : x - 3y - z = 5, \text{ and}$$

$$P_3 : 2x + 10y + 14z = 5$$

Then, which one of the following is true?

- (A) P_1 and P_2 are parallel
- (B) P_1 , P_2 and P_3 are parallel
- (C) P_1 and P_3 are parallel
- (D) P_2 and P_3 are parallel

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- (C) P_1 and P_3 are parallel
- (D) P_2 and P_3 are parallel

Solution:

$$P_1 : x + 5y + 7z = 3$$

$$P_2 : x - 3y - z = 5$$

$$P_3 : x + 5y + 7z = \frac{5}{2}$$

P_1 and P_3 are parallel as dr's of normal are same

Q79: The rate of growth of bacteria in a culture is proportional to the number of bacteria present and the bacteria count is 1000 at initial time $t = 0$. The number of bacteria is increased by 20% in 2 hours. If the population of bacteria is 2000 after $\frac{k}{\log_e \left(\frac{6}{5} \right)}$ hours, then

$\left(\frac{k}{\log_e 2} \right)^2$ is equal to

- (A) 2
- (B) 8
- (C) 16
- (D) 4

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$\left(\frac{k}{\log_e 2} \right)^2$ is equal to

(A) 2

(B) 8

(C) 16

(D) 4

Solution:

Let, number of bacteria = N

$$\frac{dN}{dt} \propto N$$

$$\frac{dN}{dt} = aN$$

$$\int \frac{dN}{N} = a \int dt$$

$$\log N = at + C$$

$$\text{At } t = 0, N = 1000 \Rightarrow C = \log 1000$$

$$\text{At } t = 2, N = 1 \cdot 2(1000) = 1200$$

$$\Rightarrow \log 1200 = 2a + \log 1000$$

$$\Rightarrow 2a = \log\left(\frac{1200}{1000}\right)$$

$$\Rightarrow a = \frac{1}{2} \log\left(\frac{6}{5}\right)$$

$$\text{At } t = \frac{k}{\log\left(\frac{6}{5}\right)}, N = 2000$$

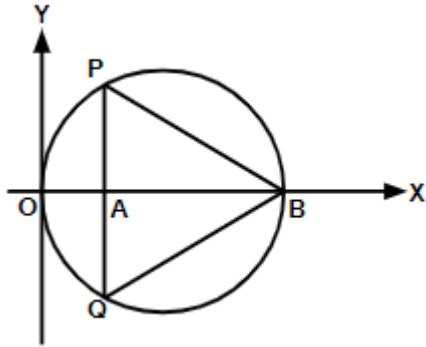
$$\Rightarrow \log 2000 = \frac{1}{2} \log\left(\frac{6}{5}\right) \cdot \frac{k}{\log\left(\frac{6}{5}\right)} + \log 1000$$

$$\Rightarrow \log\left(\frac{2000}{1000}\right) = \frac{k}{2}$$

$$\Rightarrow \frac{k}{\log 2} = 2$$

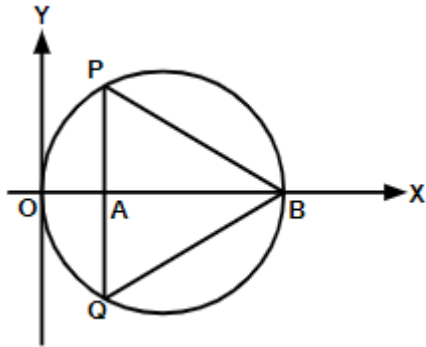
$$\Rightarrow \left(\frac{k}{\log 2}\right)^2 = 4$$

Q80: In the circle given below, let $OA = 1$ unit, $OB = 13$ unit and $PQ \perp OB$. Then, the area of the triangle PQB (in square units) is :



- (A) $24\sqrt{3}$
- (B) $26\sqrt{2}$
- (C) $26\sqrt{3}$
- (D) $24\sqrt{2}$

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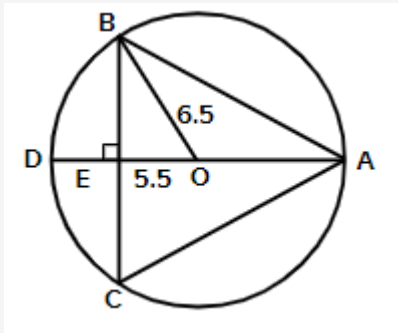
(B) $26\sqrt{2}$

(C) $26\sqrt{3}$

(D) $24\sqrt{2}$

Solution:

Let O be mid-point of AD , now perpendicular from C to BC , ($\triangle ACE$ and $\triangle ABE$ are congruent). Hence AD is diameter and O is centre of circle.



$$BE = \sqrt{(6.5)^2 - (5.5)^2} = \sqrt{12}$$

$$\text{Hence area, } \frac{1}{2} \cdot 12.2\sqrt{12} = 24\sqrt{3}$$

Q81: The number of integral values of 'k' for which the equation $3\sin x + 4\cos x = k + 1$ has a solution, $k \in \mathbb{R}$ is _____.

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11.00

Solution:

$$-\sqrt{3^2 + 4^2} \leq 3\sin x + 4\cos x \leq \sqrt{3^2 + 4^2}$$

$$-5 \leq (k + 1) \leq 5$$

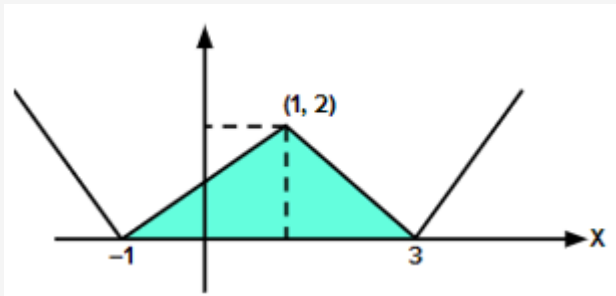
$$-6 \leq k \leq 4$$

Q82: The area bounded by the lines $y = ||x - 1| - 2|$ is _____

Q82: The area bounded by the lines $y = ||x - 1| - 2|$ is _____

4.00

Solution:



$$Area = \frac{1}{2} \times 4 \times 2 = 4$$

Q83: If $y = y(x)$ is the solution of the equation

$$e^{\sin y} \cos y \frac{dy}{dx} + e^{\sin y} \cos x = \cos x, y(0) = 0; \text{ then}$$

$$1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right) \text{ is equal to}$$

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 $e^{\sin y} \cos y \frac{dy}{dx} + e^{\sin y} \cos x = \cos x$, $y(0) = 0$; then
 $1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right)$ is equal to

1.00

Solution:

$$e^{\sin y} \cos y \frac{dy}{dx} + e^{\sin y} \cos x = \cos x$$

$$\text{Put } e^{\sin y} = t \Rightarrow e^{\sin y} \cos y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\text{So, } \frac{dt}{dx} + t \cos x = \cos x$$

$$I.F = e^{\int \cos x \, dx} = e^{\sin x}$$

Hence, solution is

$$t(e^{\sin x}) = \int \cos x \cdot e^{\sin x} \, dx$$

$$e^{\sin y} e^{\sin x} = e^{\sin x} + c$$

$$e^{\sin y} = 1 + ce^{-\sin x}$$

$$\text{Put } x = 0 \text{ \& } y = 0$$

$$\therefore \Rightarrow c = 0$$

$$\therefore e^{\sin y} = 1$$

$$\sin y = 0$$

$$\therefore y\left(\frac{\pi}{6}\right), y\left(\frac{\pi}{3}\right) \text{ \& } y\left(\frac{\pi}{4}\right) \text{ all are } 0$$

$$\therefore \text{ Required value} = 1$$

Q84: The number of solutions of the equation $\log_4 (x - 1) = \log_2 (x - 3)$ is

Q84: The number of solutions of the equation $\log_4 (x - 1) = \log_2 (x - 3)$ is

1.00

Solution:

$$\log_4 (x - 1) = \log_2 (x - 3)$$

$$\log_2 \sqrt{x - 1} = \log_2 (x - 3)$$

$$\sqrt{x - 1} = x - 3$$

$$(x - 1) = x^2 - 6x + 9$$

$$x^2 - 7x + 10 = 0$$

$$(x - 5)(x - 2) = 0$$

$$x = 5, x = 2$$

$\therefore x = 2$ is not valid

$\therefore x = 5$ the only one solution

Q85: The sum of 162^{th} power of the roots of the equation $x^3 - 2x^2 + 2x - 1 = 0$ is

Q85: The sum of 162^{th} power of the roots of the equation $x^3 - 2x^2 + 2x - 1 = 0$ is

Solution:

Let the roots be α, β, γ

$$\alpha = 1, \beta = -\omega, \gamma = -\omega^2$$

$$\text{Required value} = 1 + \omega^{162} + (\omega^2)^{162} = 3$$

Q86: The value of the integral $\int_0^{\pi} |\sin 2x| \, dx$ is _____.

Q86: The value of the integral $\int_0^{\pi} |\sin 2x| dx$ is _____.

2.00

Solution:

$$\int_0^{\pi} |\sin 2x| dx$$

Here $f(2a - x) = f(x)$

$$= 2 \int_0^{\frac{\pi}{2}} (\sin 2x) dx$$

$$= 2 \left(-\frac{\cos 2x}{2} \right)_0^{\frac{\pi}{2}}$$

$$= 2$$

Q87: Let $(\lambda, 2, 1)$ be a point on the plane which passes through the point $(4, -2, 2)$. If the plane is perpendicular to the line joining the points $(-2, -21, 29)$ and $(-1, -16, 23)$, then $\left(\frac{\lambda}{11}\right)^2 - \frac{4\lambda}{11} - 4$ is equal to _____ .

Q87: Let $(\lambda, 2, 1)$ be a point on the plane which passes through the point $(4, -2, 2)$. If the plane is perpendicular to the line joining the points $(-2, -21, 29)$ and $(-1, -16, 23)$, then $\left(\frac{\lambda}{11}\right)^2 - \frac{4\lambda}{11} - 4$ is equal to _____ .

8.00

Solution:

Direction ratio of normal to the plane is $(-1, -5, 6)$ & plane is passing through $(4, -2, 2)$ Equation of plane is

$$x + 5y - 6z + 18 = 0$$

Now, $(\lambda, 2, 1)$ is point on the plane

$$\therefore \lambda + 10 - 6 + 18 = 0$$

$$\lambda = -22$$

$\therefore$ Req. value

$$\left(\frac{\lambda}{11}\right)^2 - \frac{4\lambda}{11} - 4 = \left(\frac{-22}{11}\right)^2 - \frac{4(-22)}{11} - 4 = 8$$

Q88: Let $m, n \in \mathbb{N}$ and $\gcd(2, n) = 1$. If

$$30 \binom{30}{0} + 29 \binom{30}{1} + \dots + 2 \binom{30}{28} + 1 \binom{30}{29} = n \cdot 2^m, \text{ then } n + m \text{ is equal to}$$

_____. (Here $\binom{n}{k} = {}^nC_k$)

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$$30 \binom{30}{0} + 29 \binom{30}{1} + \dots + 2 \binom{30}{28} + 1 \binom{30}{29} = n \cdot 2^m, \text{ then } n + m \text{ is equal to}$$

_____. (Here $\binom{n}{k} = {}^nC_k$)

45.00

Solution:

$$\text{General term} = (30 - r) \cdot {}^{30}C_r$$

$$L.H.S = \sum_{r=0}^{30} (30 - r) \cdot {}^{30}C_r$$

$$= 30 \sum_{r=0}^{30} {}^{30}C_r - \sum_{r=0}^{30} r \cdot {}^{30}C_r$$

$$= 30 \cdot 2^{30} - 30 \cdot 2^{29}$$

$$= 30 \cdot 2^{29}$$

$$= 15 \cdot 2^{30}$$

$$\text{So, } n = 15, m = 30$$

$$m + n = 45$$

Q89: If $\sqrt{3} (\cos^2 x) = (\sqrt{3} - 1) \cos x + 1$, the number of solutions of the given equation when $x \in [0, \frac{\pi}{2}]$ is _____ .

Q89: If $\sqrt{3} (\cos^2 x) = (\sqrt{3} - 1) \cos x + 1$, the number of solutions of the given equation when $x \in [0, \frac{\pi}{2}]$ is _____ .

1.00

Solution:

$$\sqrt{3}\cos^2 x - (\sqrt{3} - 1) \cos x - 1 = 0$$

$$\begin{aligned}\cos x &= \frac{(\sqrt{3}-1) \pm \sqrt{(\sqrt{3}-1)^2 + 4\sqrt{3}}}{2\sqrt{3}} \\ &= \frac{(\sqrt{3}-1) \pm \sqrt{4+2\sqrt{3}}}{2\sqrt{3}} = \frac{(\sqrt{3}-1) \pm (\sqrt{3}+1)}{2\sqrt{3}} \\ &= 1, \frac{-1}{\sqrt{3}}\end{aligned}$$

Since $x \in [0, \frac{\pi}{2}]$

$$\Rightarrow \cos x = \frac{-1}{\sqrt{3}}, \text{ not possible}$$

$$\therefore \cos x = 1$$

$$\Rightarrow x = 0$$

$\therefore$ Number of solution 1

Q90: The difference between degree and order of a differential equation that represents the family of curves given by $y^2 = a \left(x + \frac{\sqrt{a}}{2} \right)$, $a > 0$ is _____.

Q90: The difference between degree and order of a differential equation that represents the family of curves given by $y^2 = a \left(x + \frac{\sqrt{a}}{2} \right)$, $a > 0$ is _____.

2.00

Solution:

$$y^2 = a \left(x + \frac{\sqrt{a}}{2} \right)$$

$$2y \frac{dy}{dx} = a$$

$$y^2 = 2y \frac{dy}{dx} \left(x + \frac{\sqrt{2y \frac{dy}{dx}}}{2} \right)$$

$$y = 2 \left(\frac{dy}{dx} \right) \left(x + \frac{\sqrt{y \frac{dy}{dx}}}{\sqrt{2}} \right)$$

$$y - 2x \left(\frac{dy}{dx} \right) = \sqrt{2} \left(\frac{dy}{dx} \right)^{\frac{3}{2}} \sqrt{y}$$

$$(y - 2y_1 x)^2 = 2(y_1)^3 \cdot y$$

Oder = 1 degree = 3

Difference = 2